The Rise of H-2A: Causal Evidence on Guestworker Substitution for Settled Farm Labor *

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Abstract

Are US farms switching to hiring agricultural labor through legal channels amid rising immigration crackdown? Non-citizen immigrant workers, two thirds of whom lack work authorization, have long supplied most seasonal farm labor in the US. In recent times, immigration enforcement has contributed to a decline in this workforce. Simultaneously, the H-2A guestworker visa program that allows farms to legally hire foreign workers has rapidly expanded. This paper asks whether the decline in settled non-citizen farm workers contributed to the rise of the H-2A program, and at what rate. To address endogeneity, I instrument the contemporaneous settled non-citizen share of farm labor with its 2000 baseline share interacted with national enforcement shifts. Farms hire about one H-2A worker for each settled worker lost that is 0.7 of the lost labor in full-time equivalents. The substitution accounts for 29 percent of aggregate H-2A growth since 2008. The effect appears only in crops, where the H-2A program is available, and not in year-round livestock. Farms are moving toward hiring through costlier legal channels with shorter contracts which has implications for agriculture and immigration policies.

JEL codes: J43, J61, J68, K37, Q12.

Keywords: H-2A program; guestworkers; immigration enforcement; farm labor; shift-share instruments; substitution.

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1 Introduction

The U.S. hired farm workforce can be divided into three parts: domestic workers, settled non-citizen workers, and seasonal guestworkers hired through the H-2A visa program. Domestic workers comprise U.S.-born and naturalized citizens, and their numbers have remained largely stable. Settled non-citizen workers, who have long been the largest source of seasonal farm labor, are workers who have established residence in the U.S., and two thirds of them lack work authorization. Their numbers have declined since 2015. Finally, H-2A guestworkers do the same seasonal farm work as settled non-citizen labor, with the difference that they are hired legally and for a short period, averaging 5.7 months. H-2A certifications grew from about 87,000 positions in 2008 to 385,000 in 2024, an increase of more than fourfold. Figure 1 depicts these trends: H-2A rises and settled non-citizen labor falls while the total remains stable. The two series measure different things and are not comparable in levels. H-2A positions are annual certifications, each covering under six months of work, while the ACS series count workers employed at the time of the survey.

This paper asks whether the decline in settled farm workers contributed to the rise of the H-2A program, and the answer is not straightforward. The two series can move in opposite directions without one replacing the other. The H-2A program can expand independently of the decline in settled non-citizen labor simply because the program becomes easier to use or because labor demand increases. Similarly, the settled non-citizen workforce can decline independently of the availability of H-2A workers, for example because fewer young workers migrate from rural Mexico. Separating substitution from these alternatives gives the rate at which H-2A guestworkers replace settled non-citizen workers, a number that matters for agricultural policy.

The substitution rate matters for three reasons. First, it informs policy aimed at assuring adequate farm labor: whether the H-2A program can supply agricultural labor when the settled workforce contracts. This has been an ongoing topic of debate given enforcement pressures (Martin and Rutledge, 2026). Second, H-2A workers are generally more expensive than settled non-citizen labor (Simnitt and Onel, 2023), because employers must pay them a wage floor[1] and provide housing and transportation, which settled workers do not require. A substitution rate near one therefore implies that farms pay more than before for the same work.[2] Third, the rate of substitution shapes how farms adjust to the loss of settled workers.

[1] H-2A employers must pay at least the highest of the Adverse Effect Wage Rate (AEWR), the prevailing wage, and the federal or state minimum wage, and they must pay the same rate to U.S. workers doing the same work. A farm that does not use H-2A faces only the minimum wage.

[2] The wage floor has been rewritten and litigated twice since 2025, after the sample period ends (Congressional Research Service, 2025). When the Department of Labor issued the October 2025 rule, it projected that the lower wage floor would bring about 119,000 additional H-2A workers (U.S. Department of Labor,

Given the fixed costs of hiring through H-2A, small farms may find the program difficult to use (Castillo et al., 2024) and fall back on consolidation or mechanization (Clemens et al., 2018; Hamilton et al., 2022).

To find the substitution rate, I use a panel of 740 commuting zones from 2008 to 2024. The regressor of interest is the settled non-citizen share of agricultural laborers. I instrument that share with each zone's settled non-citizen share in 2000 interacted with two national enforcement series, the real Border Patrol budget and border encounters during the agricultural season. Commuting zones that had a higher share of settled non-citizen workers in 2000 are affected more by enforcement. The predicted gap between high- and low-exposure zones grows in years when national enforcement is higher.

I find that farms hire approximately one H-2A worker for each settled non-citizen worker they lose. This estimate is based on the Border Patrol budget instrument.[3] Estimates using the encounter and combined instruments are 1.6 and 1.2, respectively. Since H-2A contracts are typically shorter than the work year of settled non-citizen workers, this replaces 0.7 of the lost labor in full-time equivalents. Substitution accounts for 29 percent of national H-2A growth between 2008 and 2024 with the budget instrument, and between 34 and 46 percent with the other two instruments. Event studies demonstrate that high and low exposure zones were on similar paths before enforcement intensified. Settled workers in exposed zones began to decline faster than in less exposed zones in 2015, followed by a rise in H-2A hiring from 2019. This timing is consistent with substitution. The effect is observed only in crop production, where the H-2A program is available, and not in livestock. Other tests show that the effect is identified from settled non-citizen farm labor exposure, not by general immigrant density or proximity to the border.

Several papers study what drives H-2A growth. Castillo and Charlton (2023) argue that the housing boom of the 2000s pulled workers out of agriculture and into construction, which led to higher H-2A employment within commuting zones. Arteaga and Shenoy (2024) find that a decline in Mexican migration, measured by new arrivals rather than settled workers, raises wages and H-2A certifications. Castillo et al. (2023) attribute the growth to declining domestic farm employment, which includes both citizens and non-citizens; their estimates show eighty new H-2A positions certified for every hundred domestic employees lost, over a sample ending in 2017. I measure the decline among settled non-citizens only, the group most affected by enforcement, since a majority of them lack work authorization. Over this

Employment and Training Administration, 2025).

[3]The budget is determined by Congress and not by migration flows; therefore, it has the simplest exogeneity argument of the three instruments. Additionally, it gives the smallest estimates of both the substitution rate and the percentage of H-2A growth that can be explained by substitution; thus, the benchmark is the most conservative.

paper's sample period, citizen farm employment rose rather than fell in the places that lost settled workers, so the substitution documented here is between H-2A and settled non-citizen workers rather than between foreign and domestic workers. Extending the sample through 2024 also covers the years in which most of the program's growth occurred.

Earlier studies find that immigration enforcement has reduced the immigrant workforce on U.S. farms (Martin and Rutledge, 2026) (Rutledge and Win, 2026). The effects of interior enforcement programs like 287(g) agreements and E-Verify mandates have been shown to reduce immigrant farm labor and raise farm costs (Kostandini et al., 2014; Charlton and Kostandini, 2021; Ifft and Jodlowski, 2022; Luo and Kostandini, 2022; East et al., 2023). Immigration arrests reduce non-citizen farm labor supply immediately (Gutiérrez-Li and Rubalcaba, 2025). The closest study in method is Amuedo-Dorantes et al. (2021), which finds that interior enforcement raises firms' demand for H-2B non-agricultural guestworkers. They regress H-2B requests on a county index of local enforcement policies, whereas I instrument the settled farm workforce with national enforcement series interacted with baseline exposure, following the shift-share literature (Goldsmith-Pinkham et al., 2020; Borusyak et al., 2025).

U.S. farms have experienced shortages of farm labor before. When the Bracero program ended in 1964, domestic workers did not replace the excluded Mexican workers; instead, farms mechanized or changed crops (Clemens et al., 2018). Even now domestic applicants for farm jobs are nearly nonexistent (Clemens, 2022), and farms adjust on other margins, through mechanization, crop choice, and imports (Rutledge and Mérel, 2023; Hamilton et al., 2022; Charlton, 2025). This paper shows another way farms adjust, by hiring through legal channels like the H-2A program.

The rest of the paper proceeds as follows. Section 2 describes the program, the workforce, and the data. Section 3 sets out the specification and the instruments. Section 4 reports the main estimates, the event studies, the aggregate attribution, and inference for the substitution rate. Section 5 tests the identifying assumption one threat at a time. Section 7 concludes.

2 Background and Data

2.1 The H-2A program

The Immigration Reform and Control Act (IRCA) of 1986 created the H-2A visa program and also made it unlawful to knowingly hire unauthorized workers on farms. The program allows agricultural employers to hire foreign farm workers for temporary or seasonal jobs, meaning jobs that last less than a year. Year-round operations such as dairies do not qualify. To hire through this program, an employer must demonstrate a failed attempt to recruit

domestic workers. They are required to pay the Adverse Effect Wage Rate, a regionally set wage floor above the local market rate, provide housing that passes inspection, and cover transportation to and from the worker's home country.

These requirements increase costs and cause delays in obtaining workers. The high fixed cost of participating in the H-2A program discourages small farms from using the program when they only need a few seasonal workers. As a result, the majority of H-2A use occurs on large farms and by farm labor contractors who recruit H-2A workers and provide them to multiple growers (Castillo et al., 2024; Rutledge et al., 2026). Two implications of this are important for the study.

First, since entry into the program is slow and costly, a farm that loses workers cannot quickly replace those workers through the program, especially if the farm has never used the program before.

Second, since the program does not allow for year-round employment, only farms with seasonal jobs can use the program to replace workers. Workers do not qualify for the program on their own; instead, they must be sponsored by an employer. To sponsor workers, an employer must first obtain a temporary labor certification from the Department of Labor demonstrating that there are insufficient U.S. workers available and that hiring the workers will not negatively impact the wages of similarly employed U.S. workers. Once the certification is obtained, the employer may petition U.S. Citizenship and Immigration Services for the workers. After obtaining visas from their home countries, the workers may enter the United States. In general, petitions are only approvable for workers who are nationals of countries included on a list published annually by the Department of Homeland Security.4 Workers must also meet the requirements of immigration law: unlawful presence in the United States for more than 180 days results in a three-year bar on re-entry; unlawful presence for more than one year results in a ten-year bar; and a previous removal order results in its own bar.5 Therefore, most settled workers who have been removed by law enforcement are generally ineligible to return through the program for years after being removed. The workers provided by H-2A are different people than the workers removed by law enforcement.

Although adoption was very low during the first twenty years of existence, it increased significantly over the sample period as shown in Figure 1. Growth has occurred in every state; however, the largest growth has occurred in the fruit, vegetable, and nursery regions of California, the Pacific Northwest, the Southeast, and Florida.

2.2 The settled non-citizen workforce

After the end of the Bracero Program in 1964, U.S. agriculture relied upon unauthorized migration from Mexico through migration networks and limited enforcement (Massey et al., 2016; Hanson, 2006). Many first migrated seasonally to the same region and often the same employer each year. As border enforcement increased the cost of crossing, immigrant workers settled in the United States and became the core of the seasonal workforce in labor-intensive crops (Massey et al., 2016). This workforce is now shrinking due to two factors. One factor is demographic: rural Mexico has been transitioning out of farm work for twenty years and the number of new migrants has decreased accordingly (Fan et al., 2015; Hertz and Zahniser, 2013; Charlton et al., 2019). Another factor is policy: interior and border enforcement increased during the sample period, and a large body of literature establishes that enforcement reduces immigrant farm labor supply at the local level (Kostandini et al., 2014; Charlton and Kostandini, 2021; Ifft and Jodlowski, 2022; Luo and Kostandini, 2022; Gutiérrez-Li and Rubalcaba, 2025).

Legal status is not recorded in any available data set with geographic detail. The National Agricultural Workers Survey (NAWS) is the only U.S. survey that collects self-reported status but is not identified below the regional level. Therefore, I construct the settled non-citizen agriculture worker from ACS as the observable proxy for unauthorized agricultural worker. In the fiscal 2021–2022 NAWS microdata, settled non-citizens make up 57.7 percent of hired crop workers, and 66.8 percent of them are unauthorized (U.S. Department of Labor, Employment and Training Administration, 2023). Approximately, two-thirds of the population identified by my measure is affected by enforcement. Since the non-citizen settled worker variable does not include workers who resided outside of the U.S. in the year prior to the ACS, H-2A visa holders are excluded. Appendix E contains the breakdown of the sample used for analysis and the diagnostic used to verify the exclusion. ACS may under-count settled workforce due to the fact that unauthorized workers may be reluctant to participate in the survey. The effect of an under-count on the substitution rate will depend upon whether the under-count remains constant or increases as enforcement increases. If the ACS consistently under-counts the same proportion of settled workers each year, then the measured decrease in employment will be less than the true decrease in employment and the substitution rate will overstate replacement.

On the other hand, if workers become increasingly reluctant to participate in surveys as enforcement increases, then some of the measured decrease in employment will result from workers failing to participate in surveys rather than leaving their jobs and the substitution rate will understate replacement. Under these circumstances, use of an instrument could also lead to a decrease in employment without any worker actually leaving their job and it is not

possible to differentiate between these two possibilities.

2.3 Data and Sample

The panel includes 740 commuting zones (as defined by the 1990 delineation used by Autor et al. (2013)) from 2008 to 2024, except 2020 because the Census Bureau did not release a standard one-year American Community Survey (ACS) that year.[4] County and PUMA data are aggregated to commuting zones using Dorn (2009) crosswalks and extended to the 2020 PUMA delineation through Geocorr (Appendix A). Of the 11,840 commuting zone-year cells, 11,672 have a non-missing settled non-citizen share and make up the estimation sample.

The dependent variable is the number of H-2A positions certified per commuting zone and year from the Department of Labor's Office of Foreign Labor Certification disclosure files. Each record is an employer's application for H-2A workers; it contains the number of workers requested and certified, the requested employment dates, and the worksite location. I use case numbers to identify duplicate applications and then group those applications into commuting zones based on the county where the worksite is located. Certifications are the most commonly used measure in this area of research (Castillo and Charlton, 2023) since they are available annually, can be precisely identified geographically, and directly reflect employer demand.

The main regressor is the settled non-citizen share of agricultural laborers (in percent) from the ACS. An agricultural laborer is an employed individual aged 16 to 64 working in an agricultural industry and a hired farm-labor occupation; this excludes operators, managers, and clerical workers and matches the hired workforce covered by NAWS. A farm worker is considered a settled non-citizen if they are not a U.S. citizen and were not living abroad one year prior. I refer to this measure as PANCAL, for prior-arrival non-citizen agricultural labor.

Each zone's exposure is its settled non-citizen share of agricultural laborers in 2000 from the 5% sample of the decennial census with the same industry, occupation, and citizenship filters.[5] The national shifters are the log real Border Patrol budget and the log number of U.S.-Mexico border encounters between March and September, the months in which H-2A requested start dates tend to concentrate. The crop control is the vegetable share of cropland from the 2002 Census of Agriculture.

Table 1 presents summary statistics for the estimation sample and for commuting zones above and below the median 2000 exposure. High-exposure zones employ approximately

[4]The Census Bureau released one-year ACS estimates for 2020 as experimental data due to low response rates caused by the COVID-19 pandemic. I excluded the year rather than combine the experimental file with the standard releases for consistency.

[5]The 2000 census does not ask where a person lived one year prior; I define settled workers based on years in the United States instead.

four times as many H-2A workers as low-exposure zones and more than three times as many agricultural laborers. High-exposure zones also have larger Hispanic populations and vegetable cropland shares and are located closer to the U.S.-Mexico border. These differences motivate the crop controls and testing whether the effect is driven by farm labor rather than immigrant density or proximity to the border.

3 Empirical Strategy

This paper asks how many H-2A workers are hired, on average, for every PANCAL worker lost. The simplest estimator to answer this question in the absence of endogeneity is ordinary least squares (OLS). This section explains why OLS is expected to fail and sets out the strategy used that identifies the causal effect.

3.1 Specification

The estimating equation is

$$\text{H2A}_{it} = \delta \, \text{PANCAL}_{it} + \mathbf{X}'_{it}\beta + \gamma_i + \lambda_t + \varepsilon_{it}, \tag{1}$$

where i indexes commuting zones and t indexes years. H2A_{it} is the total number of H-2A certifications, and PANCAL_{it} is the percent of settled non-citizen agricultural workers. γ_i and λ_t are commuting zone and year fixed effects, respectively. $\mathbf{X}_{it}$ is the 2002 vegetable share of cropland interacted with year dummies. The regression is weighted by year-2000 agricultural employment, so that δ measures the effect per unit of farm labor, and standard errors are clustered by commuting zone.

The parameter of interest is δ, which captures the change in the number of H-2A workers hired for a one percentage-point change in the PANCAL share. The sign is expected to be negative if farms substitute H-2A workers for settled non-citizen agricultural workers. For ease of interpretation, δ is converted into headcount and full-time-equivalent substitution rates.

3.2 Why ordinary least squares is expected to fail

I discuss three main sources of endogeneity in the OLS estimation of equation (1). First, there may be an omitted variable bias due to local labor demand shocks. While the crop controls included in $\mathbf{X}_{it}$ will capture the portion of labor demand that is related to the initial crop composition, labor demand will vary based on factors such as whether growers increase

their total acreage, switch to more labor intensive crops, open a packing facility near the area, experience a drought, etc. Therefore, the error term can be expressed as $\varepsilon_{it} = \theta D_{it} + \nu_{it}$, where D_{it} represents the local labor demand shock that is not captured by the controls. A positive demand shock increases both H-2A hiring ($\theta > 0$) and attracts or retains settled workers which gives $\text{cov}(\text{PANCAL}_{it}, D_{it}) > 0$. The omitted-variable formula gives:

$$\text{plim}\, \hat{\delta}^{OLS} = \delta + \theta \, \frac{\text{cov}(\text{PANCAL}_{it}, D_{it})}{\text{var}(\text{PANCAL}_{it})} > \delta.$$

Thus, if $\delta < 0$, then the OLS estimate will be attenuated toward zero and may potentially become positive.

Second, endogeneity may arise from measurement error in the independent variable. Classical measurement error causes attenuation of the coefficient toward zero. Most commuting zones have very small sample sizes in the ACS. Therefore, the year-to-year variation in the PANCAL share within a commuting zone can be due to sampling error.

Finally, reverse causality causes δ to be more negative. For instance, if the increase in H-2A hiring causes PANCAL workers to leave (e.g., because farmers now have access to legally available replacement workers), then the estimate will indicate substitution with the causal relationship reversed, resulting in the OLS estimate being pushed away from zero.

3.3 Instruments

I instrument the settled share with two shift-share instruments by interacting a fixed measure of each commuting zone's exposure to national enforcement series. I construct the Bartik instruments as follows:

$$(2) \qquad B_{it}^{BP} = s_{i,2000} \times \ln(\text{BP budget}_t), \qquad B_{it}^{enc} = s_{i,2000} \times \ln(\text{ag-season encounters}_t),$$

where $s_{i,2000}$ is the exposure share, the baseline PANCAL share of agricultural laborers in commuting zone i in 2000. It is constructed from the decennial census, eight years before the panel begins. The two shift components are national series that capture enforcement pressure each year: the real Border Patrol budget and the number of agricultural-season border encounters.

The logic behind these instruments is that if enforcement removes settled non-citizen workers, the effect should be larger in places that depend more on these workers. Commuting zones with high baseline settled worker shares would see a differentially larger decline in contemporaneous PANCAL share when the budget rises or border activity intensifies relative to low share zones. The instrument predicts the decline from baseline exposure and national

conditions alone, so the zone's own H-2A demand cannot change it.

The two shifters differ in character. The budget grows monotonically and captures the long-run expansion of enforcement capacity. On the other hand, the encounter series is non-monotonic and captures border activity in the months when farms hire. The budget instrument is the benchmark throughout the paper because its exogeneity argument is the simplest: federal appropriations are set nationally and unaffected by any commuting zone's farm labor market. The two instruments use largely distinct identifying variation and identify the same parameter.

3.4 Identifying assumption

Each instrument interacts one exposure share with one national time series; the overidentified specification uses both. Identification comes from the baseline exposure shares, following the exogenous-shares framework of Goldsmith-Pinkham et al. (2020). The shifters put higher weight on years with more enforcement pressure, but they do not supply independent variation. The identifying assumption is therefore that, conditional on the fixed effects and the crop controls, commuting zones with higher year-2000 exposure would have followed the same H-2A path as less exposed zones had enforcement not intensified. Equivalently, the instrument affects H-2A certifications only through the contemporaneous settled share.

Three features of the instrument design support this assumption. First, the exposure share predates the H-2A expansion as well as the enforcement buildup. I use the year 2000 for the exposure share because it allows me to use the decennial census, which covers a 5 percent sample, and because it predates movement in both the outcome and the settled share, so the variation can be argued to be exogenous. Second, the exposure share is specific to settled non-citizen agricultural labor rather than the general immigrant population or domestic agricultural workers. This enables me to focus on the specific population expected to move due to enforcement and the one the H-2A program can replace. Immigrant density outside agriculture and proximity to the border do not produce the same effect, and the effect is absent in sectors where H-2A is unavailable. Third, the event studies show that low and high baseline PANCAL exposure commuting zones are on similar paths before enforcement kicks in, and that the H-2A program responds to the decline in the PANCAL share rather than preceding it. The one pre-trend that does exist, in domestic employment, does not affect the estimate once controlled for.

3.5 First stage, reduced form, and event studies

The first stage regresses the PANCAL share on the instruments with the same controls and fixed effects,

$$\text{PANCAL}_{it} = \pi^{BP} B_{it}^{BP} + \pi^{enc} B_{it}^{enc} + \mathbf{X}_{it}'\beta + \gamma_i + \lambda_t + u_{it}, \tag{3}$$

and the reduced form regresses H-2A certifications on the instruments directly,

$$\text{H2A}_{it} = \rho^{BP} B_{it}^{BP} + \rho^{enc} B_{it}^{enc} + \mathbf{X}_{it}'\beta + \gamma_i + \lambda_t + e_{it}. \tag{4}$$

The second stage is equation (1) with the PANCAL share replaced by its fitted value from the first stage. $\hat{\delta} = \hat{\rho}/\hat{\pi}$ for just-identified regressions. Table 2 reports all three stages, the Kleibergen-Paap F-statistic for instrument strength, and the Anderson-Rubin test, which remains valid when the first stage is weak. Under homogeneous effects, the Hansen J test checks whether the two instruments identify the same underlying parameter.

To trace the dynamic response, I replace the enforcement shift component with a full set of year dummies:

$$Y_{it} = \sum_{k \neq 2008} \theta_k \left(s_{i,2000} \times \mathbf{1}[t = k] \right) + \mathbf{X}_{it}'\beta + \gamma_i + \lambda_t + \varepsilon_{it}. \tag{5}$$

For the first-stage event study, Y_{it} is PANCAL share. For the reduced-form event study, Y_{it} is H-2A certifications. With 2008 as the reference year, each θ_k measures how the gap between more and less exposed commuting zones differs relative to the same gap in 2008.

4 Results

4.1 Graphical evidence

Figure 2 plots the identifying variation directly. Panel (a) shows that commuting zones with a higher baseline PANCAL share saw larger declines in contemporaneous PANCAL between 2008 and 2024, with a slope of -0.18 (SE 0.04). This supports the relevance condition: places that historically had a larger share of settled workers are the ones where the contemporaneous PANCAL decline was largest. Panel (b) shows that those same commuting zones also experienced larger increases in H-2A worker certifications over the same window, with a slope of 76.4 (SE 18.1).

Figure 3 shows how the two variables look spatially across the country. Panel (a) shades

commuting zones by baseline PANCAL exposure, and panel (b) shades them by the change in H-2A workers between the 2008-2010 and 2022-2024 averages. The darkest zones in panel (b) overlap closely with the darkest baseline PANCAL exposure zones in panel (a), especially in California, Florida, and the Yakima Valley and Columbia Basin of Washington. The program's growth is highly concentrated such that the thirty zones in the top class account for two thirds of the national increase.

4.2 Main estimates

Table 2, column (1), gives the OLS coefficient $\hat{\delta}^{OLS} = -2.11$ (SE 1.61), which is economically small and statistically insignificant. The OLS coefficient is subject both to classical measurement error from the thin-cell ACS share measure and to endogeneity. Measurement error attenuates the coefficient mechanically and Table 3 quantifies the extent of each bias.

Columns (2) to (4) report the two-stage least squares estimates. Instrumenting the PANCAL share with the Border Patrol budget Bartik yields $\hat{\delta} = -156.7$ (61.3); the agricultural-season encounter Bartik yields -226.5 (65.7); the overidentified specification, which uses both, yields -178.9 (60.7). The coefficients are interpreted as change in number of H-2A workers per commuting zone for a one percentage-point change in the PANCAL share, so the negative sign means that a decline in the settled share raises H-2A hiring. For ease of interpretation I scale each coefficient by the workforce of the commuting zones that identify it. Two-stage least squares averages zone-level responses with weights that depend on each zone's baseline exposure and on how strongly the instrument moves its share, the Rotemberg weights of Goldsmith-Pinkham et al. (2020). Large, highly exposed zones dominate: California alone carries about half of the weight, and a few zones whose shares move against the national pattern receive small negative weights. The weighted average workforce is $\bar{N}^* = 15{,}426$ workers under the benchmark instrument, so a one percentage-point decline in the share represents about 154 workers, and the estimates imply a substitution rate of 1.02 on the benchmark instrument and between 1.0 and 1.6 across the three specifications. H-2A contracts, however, are not year-round. The average contract lasts 5.7 months (Appendix Figure B1), while settled workers average roughly eight months of farm work per year.[6] Accounting for this difference, the full-time-equivalent substitution rate is 0.72 on the benchmark instrument and between 0.7 and 1.2 across specifications. Farms in the zones that identify the estimate replace lost settled workers approximately one for one in heads, and replace about three quarters of the lost labor time.

[6] U.S. Department of Labor, National Agricultural Workers Survey, Public Data Tables, Table 2: hired crop workers averaged 35 weeks of farm work per year across the six survey cycles spanning 2007-2020 (35, 34, 35, 33, 35, and 39 weeks).

Panel B shows that the first stage is negative and precise for both instruments. The Kleibergen-Paap (KP) F-statistics test whether the instruments move the settled share strongly enough for standard inference; they range from 14.8 to 32.5 and lie above the 15 percent Stock-Yogo critical values. The first stage has the expected sign: when enforcement intensifies nationally, the PANCAL share falls by more in commuting zones that had more settled workers to begin with. The Anderson-Rubin p-values show that inference does not rest on first-stage strength. Panel C reports the reduced-form coefficients, which show that when enforcement intensifies, H-2A hiring rises by more in the more exposed zones. Each reduced-form coefficient equals the first stage times the second stage up to rounding ($55.8 \approx 0.356 \times 156.7$). When both instruments enter jointly, each remains individually significant, so the two enforcement series carry independent identifying variation. The Hansen J test does not reject the overidentifying restriction ($p = 0.167$), consistent with the two instruments identifying the same underlying parameter.

4.3 The gap between OLS and IV

The IV estimates are nearly 75-110 times larger than OLS. To evaluate this gap, I decompose the OLS-IV gap using an errors-in-variables (EIV) correction based on a direct estimate of the reliability of the PANCAL share. Using a Monte Carlo framework, I estimate the reliability ratio by holding fixed each commuting zone's true agricultural-laborer composition across pooled years. For each commuting zone-year cell, I redraw the ACS sample using its actual respondent count, recalculate the share, and residualize it under the main specification. Comparing the resulting pure sampling variance with the total residual variance yields an estimated reliability of $\hat{R} = 0.180$. This implies that less than 20 percent of the within-zone variation in the measured share is true signal after fixed effects and controls are added. Table 3 reports the decomposition. Dividing the OLS estimate by $\hat{R}$ produces an EIV-corrected coefficient of -11.7. A commuting-zone block bootstrap that re-estimates both the coefficient and $\hat{R}$ gives a 95 percent confidence interval of $[-28.1,\ 8.2]$. Even under a lower reliability of $R = 0.10$, the corrected OLS estimate reaches only -21.1. Measurement error therefore accounts for 6.2 percent of the gap between the OLS estimate (-2.1) and the benchmark IV estimate (-156.7); the remaining 93.8 percent reflects endogeneity. This is consistent with the omitted-variables argument in Section 3: positive local labor demand shocks raise H-2A hiring while attracting or retaining settled workers, inducing a positive correlation between the error term and the PANCAL share and biasing OLS toward zero. The instrument therefore corrects endogeneity bias rather than merely adjusting for sampling noise.

4.4 Event study dynamics

The main estimates report a single coefficient for a pattern spanning sixteen years. Figure 4 shows how the relationship between baseline exposure and each variable, the settled share and H-2A hiring, evolves over time. I interact the year-2000 PANCAL share with a set of year dummies, with year 2008 as reference. Each coefficient captures how the gap in each variable evolves in high PANCAL versus low PANCAL exposure regions relative to year 2008.

Panel (a) shows the first stage. The coefficients are close to zero through 2014 and are not jointly different from zero over 2009-2012. They become significant at the 5 percent level from 2015. This means that exposed commuting zones lost settled non-citizen workers faster than unexposed ones from 2015 onward, and the gap widened further after 2019. Panel (b) shows the reduced form. The early coefficients are small and negative, and jointly different from zero. The coefficients become positive and significant at the 5 percent level from 2019, rise through 2023, and remain large in 2024. Two features are important to note. First, if exposed commuting zones had been hiring more H-2A workers than other zones before enforcement intensified, the early coefficients would be positive. Instead they are negative, which works against the effect rather than inflating it. Second, the coefficients through 2015 are small relative to the later period, about 8 percent of the 2023 peak at most, and largest in 2009, the recession year.

The two panels together depict the substitution pattern within a timeline. The settled workforce in exposed commuting zones begins to decline in 2015, and H-2A hiring in the same zones rises significantly from 2019. The gap between the two dates describes the timing; it is not an estimated lag, and Appendix C shows that the design cannot identify the lag length directly. The reduced form also accelerates in 2019, which coincides with the larger decline in the settled worker share.

4.5 Aggregate attribution

Table 4 shows how much of the growth in H-2A employment can be explained by the PANCAL decline. Between the 2008–2010 and 2022–2024 averages, summing over the 735 commuting zones observed in both windows, H-2A employment increased by 298,923 workers, while the settled non-citizen agricultural workforce fell by 83,912. Multiplying the decline of 83,912 workers by the benchmark substitution rate gives a predicted increase of $1.02 \times 83{,}912 = 85{,}243$ H-2A workers, which accounts for 29 percent of the observed growth. For the encounter and overidentified rates the shares are 46 and 34 percent, respectively. The settled-worker decline is therefore a large contributor to the program's growth. The remaining contributors include places that adopted H-2A without a history of reliance on settled non-citizen workers;

Figure 3 shows moderate growth across the Midwest, where fruit and vegetable production is limited and H-2A use was minimal at baseline. They also include places where both the settled workforce and H-2A employment rose because local labor demand grew, variation the instrument deliberately sets aside.

4.6 Interpreting the magnitude

The main coefficients in Table 2 report cluster-robust standard errors whose confidence intervals assume that $\hat{\delta}$ is approximately normal in large samples. With a sample of 740 commuting zones, this assumption would be reasonable, but when a large share of the weight is carried by a few commuting zones, the effective sample can be far below 740. I therefore run a commuting-zone block bootstrap, which does not rely on this approximation. In each of the 999 replications, I draw the 740 commuting zones with replacement, re-estimate two-stage least squares, and form the substitution rate $|\hat{\delta}|/(0.01 \times \bar{N}^*)$. Table 5 reports the resulting percentile intervals. The denominator is held at the full-sample first-stage average, $\bar{N}^* = 15{,}426$, so the rate's uncertainty comes from $\hat{\delta}$. Appendix D reports a version that also recomputes $\bar{N}^*$ in each replication; its median is also near one, and its tails are driven by weakly identified replications. Figure 6 shows that the sampling distribution is right skewed, which is why the intervals are percentile based rather than symmetric.

Three conclusions follow from the intervals. First, the headcount rate is centered near one: the point estimate is 1.02, the bootstrap median is 1.11, and the probability that the rate falls below one is 42 percent, so the data do not separate replacement below one for one from replacement above it. Second, replacement well above one for one cannot be ruled out; the upper bound of the featured interval is 3.2. Third, the labor-time rate is centered just below one, with a median of 0.79 and an interval of $[0.20, 2.26]$, so exact one-for-one replacement of labor time cannot be rejected. A delta-method standard error that assumes normality is 0.40, well below the bootstrap standard error of 1.04, and would overstate precision considerably.

The right tail has one identifiable source, documented in Appendix D. Farm labor is geographically concentrated, and bootstrap draws that underweight the California core thin out the identifying variation, weaken the first stage, and inflate $|\hat{\delta}|$. Conditioning on replications with a first-stage F above 10 trims the upper bound from 3.2 to 2.0 and moves the probability of a rate below one from 42 to 53 percent, so weak first stages inflate the rate rather than shrink it.

5 Robustness

The causal claim of the main results holds if the identifying assumption is valid: the Bartik instrument, baseline PANCAL exposure times national enforcement, affects the outcome variable, H-2A certifications, only through changes in the main regressor, the contemporaneous PANCAL share. There is no single way of testing this, so in this section I run a number of tests, each checking one way in which the estimate could be wrong. The same tests also justify several choices made in constructing the variables and the specification: why the main regressor is the settled non-citizen share rather than a broader immigrant measure, why the crop controls are constructed the way they are, how the estimate changes when the definition of the agricultural workforce is widened or narrowed, and why two instruments are used and how they differ.

5.1 Testing the exposure measure against immigrant density and border proximity

I test whether the substitution rate proposed above picks up the decline of settled non-citizen agricultural workers or simply captures regions with higher immigrant density in general. The two are hard to tell apart. Suppose two commuting zones have the same baseline PANCAL share. The first has the majority of its immigrant workers in agriculture. In the second, agricultural workers are a small fraction of a much larger immigrant population working in non-agricultural sectors. Enforcement affects both zones' farm workers similarly, but in the second it also affects the larger immigrant workforce outside agriculture. If the second kind of place also expanded H-2A hiring when enforcement tightened, for reasons unrelated to losing farm workers, the instrument would be picking up immigrant density in general and the coefficient would overstate substitution.

To test the above, I build a non-agricultural version of the Bartik instrument in which the exposure share is the settled non-citizen share of the non-agricultural workforce, interacted with the same enforcement shifters. To make the non-agricultural Bartik comparable to the agricultural one, I restrict the non-agricultural population to low-education workers. I then enter both Bartiks in one reduced form,

$$\text{H2A}_{it} = \pi_{ag} B_{it}^{ag} + \pi_{nonag} B_{it}^{nonag} + \mathbf{X}_{it}'\beta + \gamma_i + \lambda_t + \varepsilon_{it}, \tag{6}$$

where B_{it}^{ag} is the agricultural Bartik of the main specification and $B_{it}^{nonag} = s_{i,2000}^{nonag} \times \ln(\text{shift}_t)$ is its non-agricultural counterpart. If immigrant density were what mattered, the two would split the effect or the non-agricultural Bartik would dominate. Table 6 shows the opposite.

The agricultural Bartik keeps a positive, significant coefficient, while the non-agricultural Bartik has the wrong sign and is imprecise. I also add the contemporaneous non-agricultural settled non-citizen share as a control in the main specification, and the estimate moves modestly, and away from zero, as Panel B of Table 6 shows.

Enforcement studies have used distance from the Mexican border as a variable that measures enforcement exposure. Being closer to the border would imply higher unauthorized activity and hence higher enforcement exposure. To see whether the variation actually comes from activity closer to the border, I interact negative log distance to the border with the same budget shifter as in the main specification. The second column of Table 6 shows that this instrument is weak (first-stage F of 4.7) and gives an imprecise estimate. When put in a reduced-form horse race with the share-based Bartik, the share-based Bartik survives. Thus, proximity to the border does not identify the effect once the location of settled non-citizen workers is accounted for.

One exclusion story is specific to farms. Growers in high-exposure zones could see enforcement rising and enter the H-2A program before losing any settled workers, as insurance. H-2A hiring would then rise while the PANCAL share was unchanged, and the non-agricultural comparison above would not detect it. The event studies do. Anticipation of this kind would place the H-2A response ahead of the PANCAL decline. The order in Figure 4 is the reverse. The first stage moves in 2015 and the reduced form in 2019, and the H-2A coefficients show no positive pre-trend.

5.2 Testing for a response where H-2A is unavailable

H-2A is a seasonal program, limited to jobs that last less than a year. There are sectors that employ settled non-citizen workers but may not have access to H-2A because they require year-round employment. Livestock production is such a sector: dairy, its largest employer, along with cattle, hog, and poultry operations, relies on year-round labor, so enforcement may lead to a decline in settled workers but an H-2A alternative does not exist in most cases. If an H-2A response exists where the visa is not available, that would mean my results are picking up a different channel. If the response is substitution, it should be absent in livestock.

The main instrument's exposure is a broad definition of all agricultural laborers. Here, I split it into its two sectors. I define the crop share as the settled non-citizen share of laborers in crop production and agricultural services (IPUMS 1990 industries 10 and 20) and the livestock share as the settled non-citizen share of laborers in livestock production (industry 11), for each commuting zone and year.

I ask two questions of the livestock Bartik. First, does enforcement actually remove

livestock settled workers, so that the placebo carries a real shock? The contemporaneous livestock share is the dependent variable in the first stage. Second, does that shock raise H-2A hiring? The second question is answered by letting the two components compete in one reduced form,

$$\text{H2A}_{it} = \pi_{crop} B_{it}^{crop} + \pi_{lvstk} B_{it}^{lvstk} + \mathbf{X}_{it}'\beta + \gamma_i + \lambda_t + \varepsilon_{it}. \tag{7}$$

The 2000 values of each share are interacted with the enforcement shifter to give a crop Bartik B_{it}^{crop} and a livestock Bartik B_{it}^{lvstk}, constructed exactly as the main instrument. Table 7 reports both. The first stage for livestock is negative and significant with either enforcement series, implying that enforcement drains the livestock settled workforce as it does the crop workforce and the broader agricultural workforce in the main specification. However, in the horse race, H-2A certifications respond to the crop Bartik while the coefficient on the livestock Bartik is negative and imprecise. This means that the substitution response appears only where H-2A is available.

5.3 Testing for pre-existing trends in exposed places

The substitution rate attributes the rise in H-2A hiring to the loss of settled non-citizen workers. A competing explanation attributes it to a different loss. Castillo et al. (2023) argue that H-2A growth followed a decline in domestic farm employment, and estimate that roughly eighty H-2A jobs were certified for every hundred domestic employees lost. If domestic workers in exposed commuting zones were leaving agriculture, aging out or moving into construction and services, and H-2A replaced them, then my instrument could be crediting the settled-worker channel for a channel that runs through citizens instead. The two stories have different policy implications: one says enforcement created the vacancies, the other says demographics did.

The two explanations make different predictions for domestic employment. If H-2A were replacing departing citizens, domestic employment in exposed zones should fall as enforcement tightens. If H-2A were replacing settled non-citizens, domestic employment need not fall. I run the main specification with domestic agricultural employment, meaning U.S.-born and naturalized citizen workers, in place of H-2A workers as the outcome. Panel A of Table 8 shows that domestic employment does not fall; it rises, and by a lot. Taken at face value, each settled worker lost would be matched by 1.3 to 2.5 additional domestic workers, depending on the instrument. Whatever H-2A replaced in exposed zones, it was not departing citizens. The workforce that declined is the settled non-citizen one, and citizen employment in the same places grew.

That growth raises a second question. A rising domestic workforce in exposed places suggests those places were changing in ways unrelated to enforcement, most plausibly expanding, and expansion could raise every kind of hiring, H-2A included. Figure 5 confirms the suspicion. The event study of domestic employment on baseline exposure shows a gap that opens in the very first year after the reference year and grows smoothly through the period, and the joint test that the 2009-2012 coefficients are zero rejects at the 1 percent level. This is a pre-existing trend, not a response to enforcement, and it is exactly the pattern the same test did not find for H-2A in Section 4.4. It is also not an artifact of settled workers naturalizing and being reclassified as citizens: restricting to U.S.-born workers, a group no one can join by naturalizing, gives the same pre-trend, also rejecting at the 1 percent level, and most of the coefficient.

The question that matters for the substitution rate is whether it rests on this trend. I control for it directly. Let G_i be commuting zone i's growth in domestic agricultural employment over 2008-2012, which is predetermined and only weakly correlated with exposure, and add it to the main specification interacted with year dummies,

$$\text{H2A}_{it} = \delta \, \text{PANCAL}_{it} + \sum_t \theta_t \, G_i \times \mathbf{1}[\text{year} = t] + \mathbf{X}'_{it}\beta + \gamma_i + \lambda_t + \varepsilon_{it}, \tag{8}$$

so that each commuting zone can follow its own time path scaled by its pre-trend, and δ is identified only from variation orthogonal to pre-existing domestic trajectories. Panel B of Table 8 shows that the estimate does not change qualitatively. The benchmark coefficient stays within half a standard error of its original value, and the encounter and overidentified estimates move toward it so that the two instruments agree more closely than before. The substitution estimate does not live on the domestic trend. A more aggressive control, contemporaneous domestic employment itself, is endogenous to the same shocks and is not a valid specification.

5.4 Testing whether the instruments identify the same parameter

An instrument is only as good as the argument behind it. Section 4 reported two instruments with the same baseline exposure share but different national enforcement shifters. The Border Patrol budget captures structural enforcement capacity which grows monotonically year by year while the agricultural-season encounters captures contemporaneous border movement and is non-monotonic. The two together capture different variations of the same underlying parameter (Hansen J test does not reject, $p = 0.17$). The agreement is informative because the two instruments share little of the variation the regression uses: after fixed effects and controls, the correlation of their residual variation is only 0.28. Identification comes from the exposure share and not the shifts. The comparison of instruments in this section tests

whether the timing of the H-2A response follows enforcement.

Table 9 pushes on this from two directions. Panel A shows that if the shift component of the instrument is replaced with any other monotonically increasing series like consumer price index, federal health spending or a simple linear trend, it should behave like the Border Patrol budget, and it does. This is the critique of monotone shifters raised by Goldsmith-Pinkham et al. (2020). The encounter instrument is the answer to this limitation. Appendix Figure A1 plots the different national series I use here. The benefit of using the agricultural-season border encounter shift is that its variation does not resemble other monotonic series, its movements are policy driven, and yet it gives a stronger KP F-statistic. The paper treats the budget instrument as the transparent benchmark and leans on the encounter instrument, and on the overidentification test, for the claim that the variation is enforcement rather than trend.

The literature on immigrant labor supply typically uses instruments that capture national changes in a group's supply, allocated to places where that group was already concentrated (Card, 2001). To connect the estimates of this paper to this conventional design, I construct an instrument whose shift component measures the national decline in PANCAL workers, keeping the baseline PANCAL share as the exposure variable. There are two versions of this instrument: a leave-one-out Bartik, in which the national series of the PANCAL decline excludes the own commuting zone, and another that includes it. This supply-shift instrument rests on a different argument from this paper's. It allocates the entire national decline to places by baseline shares, irrespective of the cause of that decline, rather than only its enforcement-driven part. I further run overidentification tests with the leave-one-out Bartik paired with each of the two enforcement instruments individually, and then with all three together. Panel B of Table 9 shows that the supply-shift instrument gives an estimate close to the enforcement instruments. No pairing rejects the overidentifying restriction at 5 percent. The estimate is closest to the encounter instrument's and the pairing with the budget instrument is borderline, the same gap between smooth and non-smooth shifters that Panel A documents. The substitution rate is therefore a property of losing settled workers, whatever the cause, rather than of enforcement specifically.

The tests so far show whether the instruments agree; I now turn to where their agreement comes from. Goldsmith-Pinkham et al. (2020) discuss how to interpret an overidentified IV estimate. It is a weighted average of just-identified estimates, and the weight on each measures its contribution to the first-stage covariance between the instrument and the settled share. The weights can be computed across the two instruments, across years, and across commuting zones. Table 10 reports them along all three dimensions. The budget instrument receives 68 percent of the weight and the encounter instrument 32 percent. The weighted

just-identified estimates reproduce the overidentified coefficient. The weights across years are largest for 2019, 2023, and 2024, which are the surge years. The pre-2018 years carry negative aggregate weight. This implies that in those years enforcement had not yet risen and the settled share had not yet fallen relative to the panel mean, so they act as the before period that the surge years are compared against. Negative weights of this kind are a feature of the decomposition, not a sign of misbehavior (Goldsmith-Pinkham et al., 2020). Across commuting zones, California carries 49 percent of the identifying weight on the benchmark instrument and 45 percent on the overidentified estimate. That concentration is not an accident of the design. It follows from where American farm labor is and from weighting the estimate by farm employment.

5.5 Alternative specifications and samples

Table 11 reports how the controls in the main specification were chosen. Without the crop controls, the estimates are larger and less precise. Areas with higher crop demand could hire more H-2A workers even without a substantial decline in PANCAL workers, and the crop controls account for this demand. I construct three types of crop controls, each fixed at its 2002 Census of Agriculture value and interacted with year dummies: a labor-intensive crop share, which is the combined acreage of vegetables, orchards, and berries as a share of cropland; the vegetable, orchard, and tobacco shares entered as separate controls; and the vegetable share of cropland alone. Table 11 shows that the first two absorb too much of the identifying variation and weaken the first stage. The vegetable share of cropland, on the other hand, keeps the first stage strong while absorbing the crop composition that both attracts settled workers and uses H-2A.

Section 4.6 shows that the California core anchors the substitution rate, so the natural check is to remove it. Table 12 indicates what happens to the estimates when I drop California from the sample or only the three largest zones in California. In the former case, the coefficient doubles and the first stage weakens to an F of 5.5, because most of the identifying variation leaves with the state. In the latter case, which leaves the rest of the state in, the coefficient stays within a few units of the benchmark. No rate is reported for either row. The rate divides by the workforce of the zones that identify the estimate, and without the California core that workforce falls from 14,913 to 1,065 and 3,953 workers while the first stage weakens, so the denominator is no longer stable (Appendix D). The coefficient does not depend on California; the denominator does. The unweighted specification makes the same point from the other direction. It weighs all commuting zones equally, but a percentage point decline in PANCAL in California is not comparable with the same decline in Iowa, so the coefficient

is smaller in levels and the identifying workforce smaller, and the rate, 1.9, is close to the benchmark. The weighted specification measures the effect per unit of farm labor, which is what the substitution question asks. Dropping the top one percent of commuting-zone-year H-2A outcomes leaves the coefficient unchanged while the first stage weakens.

The second group varies the sample and the timing. The crop agriculture sector covers crop production and agricultural services (IPUMS 1990 industries 10 and 20); it is the sector that employs the most PANCAL and H-2A workers. Restricting the sample to crops gives a substitution rate close to two. The PANCAL construction specifically excludes recently arrived non-citizens. If they affected my estimates, including their share as a control variable would shift the main coefficient. As Table 12 shows, the coefficient barely moves, reinforcing the group-quarters evidence in Appendix E that recent arrivals do not contaminate the settled measure. Lagging the instruments by a year, which allows enforcement time to bite, gives the same coefficient but a weaker first stage. Lagging the settled share instead, which matches H-2A applications to the labor conditions observed a year earlier, gives a slightly larger coefficient. Neither changes the estimate substantially.

The next variation widens the definition of the main regressor to all Mexican-born or all foreign-born agricultural laborers, including citizens and recent arrivals, following Castillo et al. (2023). The wider definitions give larger coefficients. These are the only two specifications in the paper where the Hansen J test rejects. This is a finding on its own. The two enforcement instruments agree when the effect is measured on settled non-citizen agricultural labor, not on the wider definitions, where the enforcement channel is diluted. The difference comes from the inclusion of citizens, who are unaffected by enforcement, and recent arrivals, who likely include H-2A workers themselves. The settled non-citizen share is the definition on which the two instruments tell the same story, which is why it is the paper's main regressor.

Splitting the panel into two halves, 2008 to 2017 and 2018 to 2024, I re-estimate the main specification in each. As noted in Section 5.4 through the Rotemberg weights, neither half has enough identifying variation to estimate $\hat{\delta}$ precisely. This is because the enforcement variation is smaller within either window. The point estimates follow the event-study timing, small in the early half and large in the late half, so the two halves read like a before and after comparison.

Finally, Table 12, Panel B, shows what happens when the outcome variable is proportional instead of in levels. The inverse hyperbolic sine of H-2A workers as the outcome gives a coefficient of -0.089 (0.038), which means that a one percentage-point decline in the PANCAL share raises H-2A employment by about 9 percent. The log of H-2A workers, dropping the 2,531 commuting-zone-years with zero H-2A (480 of them in commuting zones that never use the program), gives a coefficient of -0.124 (0.037) with a first-stage F of 14.8, which

means a 12 percent increase in H-2A employment per percentage point. The proportional transformations compress the large-H-2A zones that carry most of the identifying variation, so the implied rates, 0.59 and 0.90, are below the main specification's 1.20.

6 Conclusion

This paper estimates that farms hire about one H-2A guestworker for each settled non-citizen agricultural worker lost. Because H-2A contracts are short, this replaces about three quarters of the lost labor time. The variation comes from national immigration enforcement, which removed more settled workers in places that had more of them in 2000. Substitution for lost settled workers accounts for about 29 percent of H-2A growth between 2008 and 2024. The result survives alternative national shifts, crop composition controls, and falsification tests against non-agricultural labor markets, border proximity, and sectors where the program is unavailable. Event studies show that the settled workforce in exposed places begins to decline in 2015 and that H-2A hiring responds from 2019. The program therefore serves as an adjustment margin when the settled workforce contracts. Agriculture did not so much lose labor as change who supplies it, from settled workers living in the United States to temporary workers recruited abroad, through a channel that costs more because of its wage floor, housing, and transportation requirements.

7 Conclusion

This paper asks whether the decline in the number of settled non-citizen farmworkers contributed to the rise of the H-2A program, and if so, at what rate. For every settled worker lost, farms hire approximately one H-2A worker. Since H-2A contracts are typically short-term, this replaces approximately three quarters of the lost labor time. Replacement of lost settled workers accounts for approximately 29 percent of the total growth of the H-2A program between 2008 and 2024. Places with a larger share of settled workers in 2000 experienced a faster decline in their settled workforce from 2015 and a faster increase in H-2A hiring from 2019.

This estimate is based on comparisons of commuting zones with a higher share of settled workers in 2000 versus those with a lower share. Therefore, it represents only one component of the program's expansion: H-2A hiring that replaced settled workers who departed. Many other factors contributed to the growth of the program. For example, use of the H-2A program increased throughout the Midwest, where very few settled workers worked in agriculture in 2000. Additionally, labor demand increased in certain areas, and the program may have

become easier to use. Some positions previously filled by new migrants from Mexico may now be filled through the H-2A program. None of these factors is included in the 29 percent. California carries about half of the identifying weight; thus, the rate reflects replacement primarily in areas where the settled workforce was largest.

The results have three implications. First, the H-2A program provides a way for farms to adjust their workforce after losing settled workers. Rather than simply doing less work, farms in these regions replaced most of the lost labor time. Second, replacing settled workers is costly. H-2A employers must pay a wage floor and provide housing and transportation, neither of which settled workers need. Therefore, the same work is more expensive, and changes to the wage floor and application fees directly affect the cost of adjusting. Finally, when farms can use the H-2A program, losing settled workers mainly changes who does the work instead of how much work is done. Settled workers living in the U.S. are replaced by temporary workers recruited from other countries. Since dairies and other year-round businesses cannot participate in the H-2A program, they do not have this option.

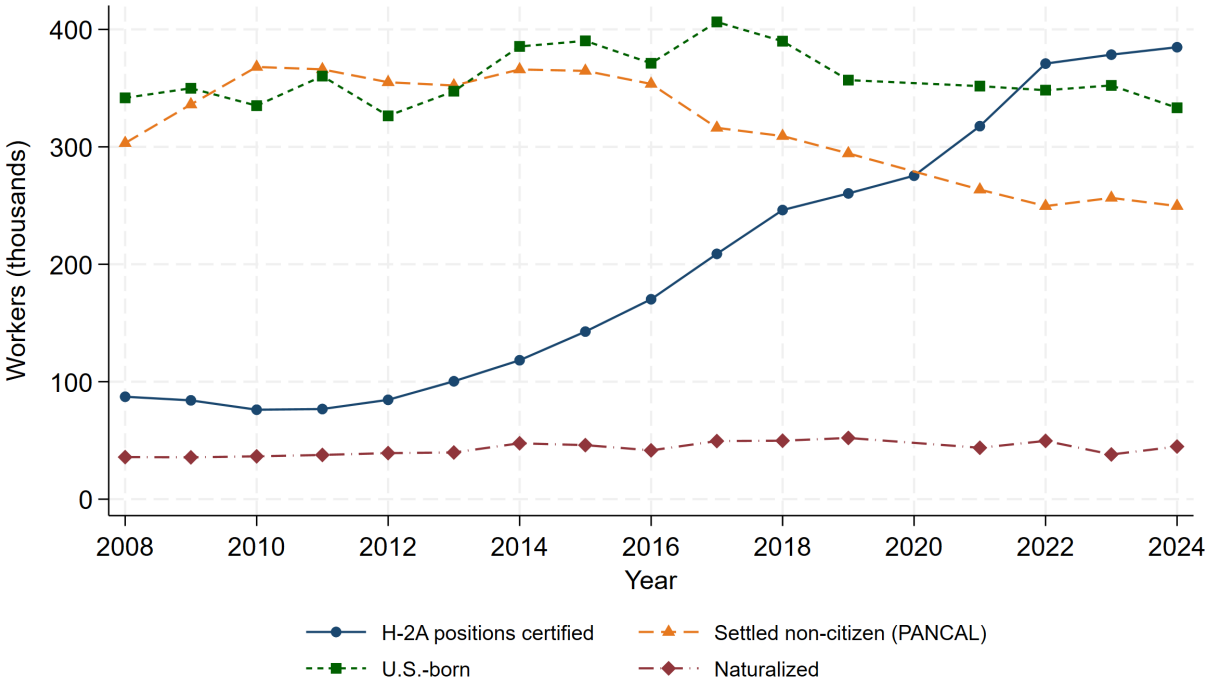


FIGURE 1. Components of the U.S. Hired Farm Workforce, 2008-2024

Notes: National totals in thousands of workers. H-2A positions certified are from the Department of Labor OFLC disclosure files, full and partial certifications, by fiscal year. The three ACS series are person-weighted counts of agricultural laborers, employed persons aged 16 to 64 in crop production, livestock production, or agricultural services in farm-worker, nursery-worker, or supervisor occupations, split by citizenship: settled non-citizen (PANCAL) workers are non-citizens who did not live abroad in the prior year, and the U.S.-born and naturalized series are citizens. The ACS did not release standard one-year estimates for 2020, so the ACS series have no value in that year. *Sources:* Department of Labor OFLC disclosure files; American Community Survey one-year files via IPUMS.

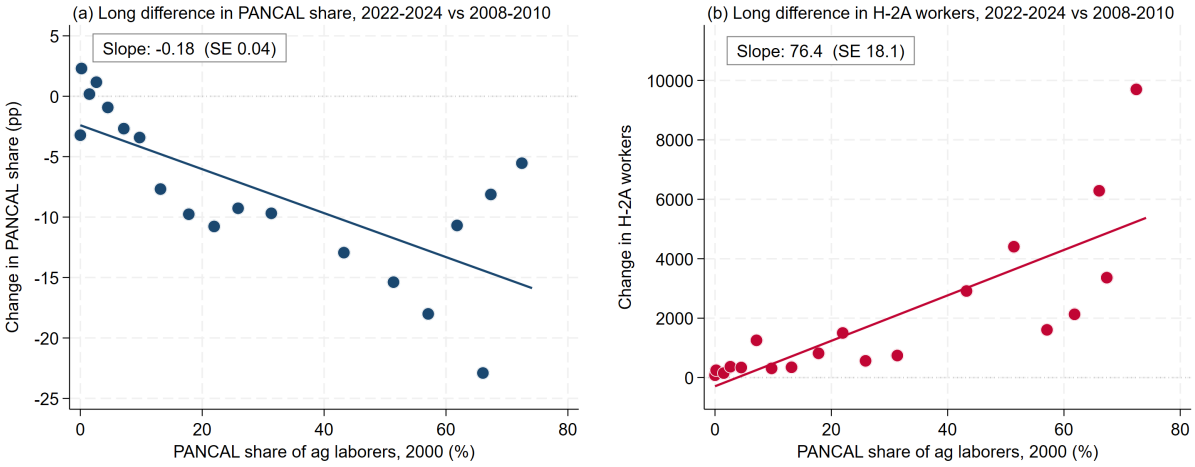


FIGURE 2. The Identifying Variation in Long Differences

Notes: Each panel plots a long difference against baseline exposure across the 740 commuting zones in the estimation sample. Baseline exposure is the PANCAL share of agricultural laborers in 2000. The long difference is the 2022–2024 average minus the 2008–2010 average, of the PANCAL share in percentage points in panel (a) and of certified H-2A workers in panel (b). Commuting zones are sorted by exposure and grouped into nineteen bins each holding approximately 5 percent of year-2000 agricultural employment; a commuting zone holding at least 5 percent by itself forms its own bin. Each dot is the employment-weighted mean of the bin. Fitted lines and the reported slopes are from regressions at the commuting-zone level, weighted by year-2000 agricultural employment, with heteroskedasticity-robust standard errors; they are not fits through the binned points. *Sources:* American Community Survey via IPUMS; Department of Labor OFLC disclosure files; Census 2000.

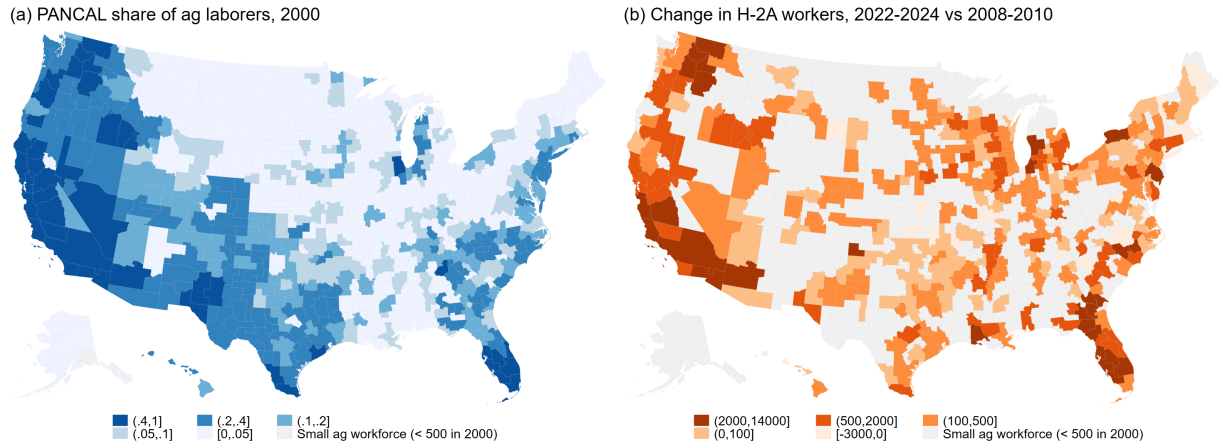


FIGURE 3. Baseline Exposure and the Growth of H-2A Employment, by Commuting Zone

Notes: Both panels show the 740 commuting zones in the estimation sample. Panel (a) shades zones by the PANCAL share of agricultural laborers in 2000, in round classes; the 10 percent boundary matches the high-exposure definition used elsewhere in the paper. Panel (b) shades zones by the change in certified H-2A workers, 2022–2024 average minus 2008–2010 average. Its class boundaries are zero, separating growth from no growth, and round values chosen so that the classes above zero hold 1.5, 11.8, 21.2, and 66.4 percent of the national increase, from lightest to darkest. Zones with fewer than 500 agricultural workers at baseline are left unshaded in both panels; the shaded zones carry 84.3 percent of year-2000 agricultural employment. *Sources:* American Community Survey via IPUMS; Census 2000; Department of Labor OFLC disclosure files; 1990 commuting zone boundaries.

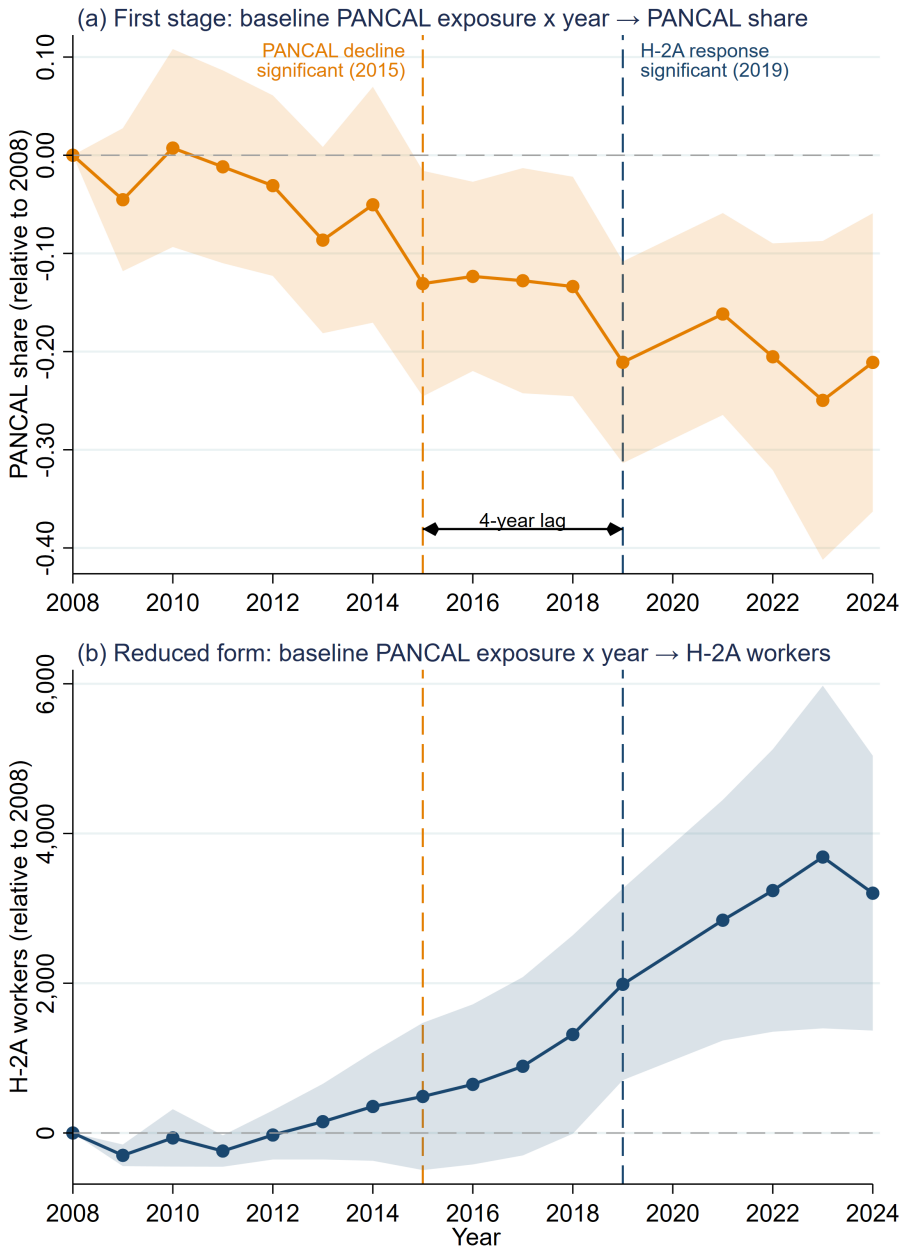


FIGURE 4. Event Studies of Baseline PANCAL Exposure

Notes: Coefficients from equation (5), with the PANCAL share as the outcome in panel (a) and H-2A workers in panel (b), estimated on the 740 commuting zones in the main sample with commuting zone and year fixed effects and vegetable share × year controls, weighted by year-2000 agricultural employment. There is no coefficient for 2020 because the ACS did not release standard one-year estimates for that year. Shaded bands are 95 percent confidence intervals clustered at the commuting zone level. Dashed vertical lines mark the first year in which the coefficient sequence is significant at the 5 percent level. *Sources:* American Community Survey via IPUMS; Department of Labor OFLC disclosure files.

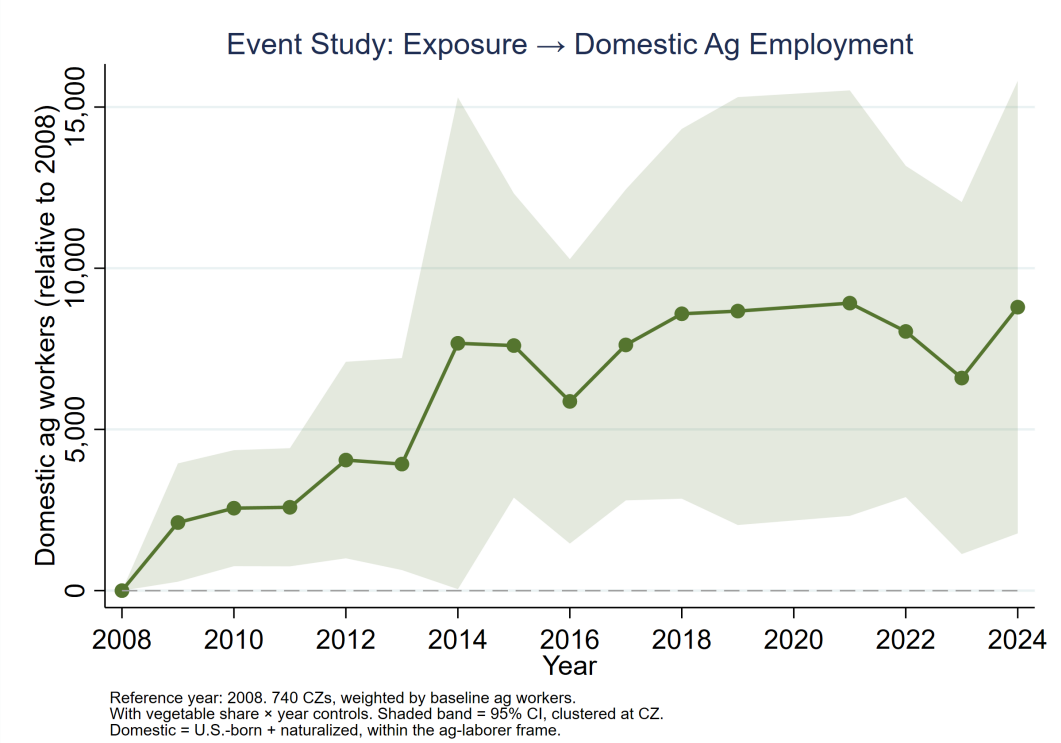


FIGURE 5. Event Study of Domestic Agricultural Employment on Baseline PANCAL Exposure

Notes: Each point is the coefficient on the year-2000 PANCAL share interacted with a year dummy, with domestic agricultural employment (U.S. citizens, born or naturalized) as the outcome and 2008 as the reference year. 740 commuting zones, weighted by year-2000 agricultural employment, with vegetable share × year controls. Shaded band is the 95 percent confidence interval clustered by commuting zone. The joint test that the 2009–2012 coefficients equal zero rejects at the 1 percent level.

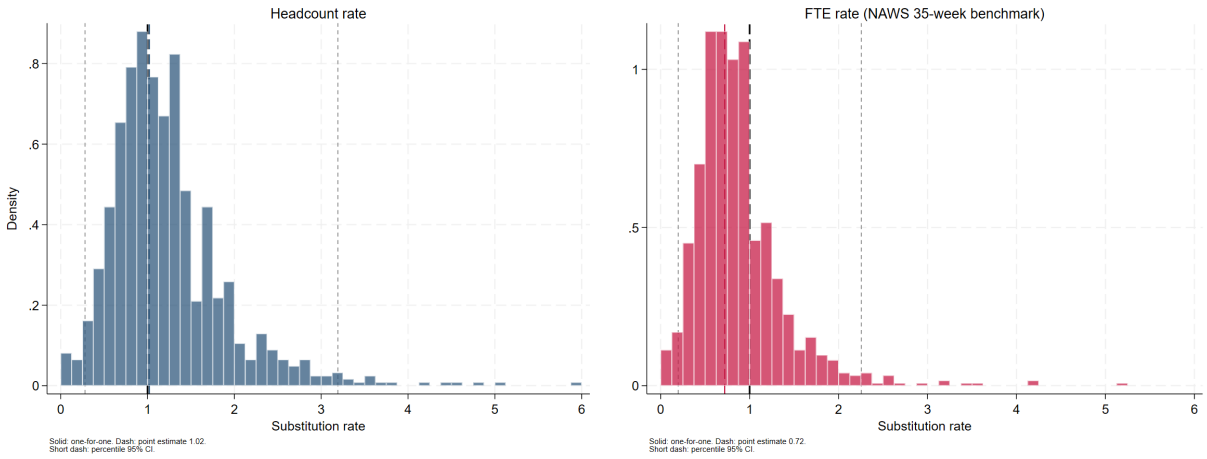


FIGURE 6. Bootstrap Distribution of the Substitution Rate

Notes: Histograms of the fixed-denominator substitution rate across 999 commuting-zone block bootstrap replications of Table 2, column (2), in headcount (left) and NAWS full-time-equivalent (right) units. The solid vertical line marks one-for-one replacement, the dashed line the full-sample point estimate, and the short-dashed lines the percentile 95 percent interval. Replications with a rate above 6 are not shown. See Table 5 and Appendix D.

TABLE 1. Summary Statistics

	(1) Full sample	(2) Low exposure (below median)	(3) High exposure (at or above median)
Panel A. Outcome and main regressor			
H-2A workers (count)	257.7	105.9	409.4
	(916.2)	(299.4)	(1,242.3)
PANCAL share (percent)	19.6	8.2	30.8
	(22.8)	(13.9)	(24.3)
Panel B. Agricultural employment and crop mix			
Total agricultural laborers	995.6	436.4	1,554.8
	(3,317.7)	(538.7)	(4,593.6)
Vegetable cropland share (percent)	1.0	0.3	1.8
	(3.4)	(0.9)	(4.6)
Panel C. Demographics and labor market			
Working-age population	280,794	128,718	432,870
	(779,586)	(281,307)	(1,044,134)
Hispanic share (percent)	11.4	4.5	18.3
	(14.7)	(3.7)	(18.0)
Employment rate (percent)	68.1	69.2	66.9
	(7.0)	(7.6)	(6.2)
Panel D. Geography			
Distance to U.S.-Mexico border (km)	1,503	1,726	1,281
	(763)	(716)	(744)
Panel E. Baseline (year-2000) shares, percent			
PANCAL share, 2000	12.0	1.4	22.7
	(14.7)	(1.8)	(14.2)
Non-agricultural settled non-citizen share, 2000	3.3	1.3	5.4
	(4.4)	(1.2)	(5.3)
Agricultural laborers, regression-weighted mean	9,062.5	—	—
Agricultural laborers, first-stage weighted mean ($\bar{N}^*$)	15,426.0	—	—
Observations (commuting zone-years)	11,840	5,920	5,920
Commuting zones	740	370	370

Notes: Panel means with standard deviations in parentheses; all shares and rates in percent. Column (1) is the full panel, 740 commuting zones over 2008 to 2024 excluding 2020; cells with zero agricultural employment have a missing PANCAL share and are dropped from estimation. Columns (2) and (3) split commuting zones at the median year-2000 PANCAL share of 6.0 percent. H-2A workers are certified positions, full and partial certifications. The full-sample mean of the 2000 PANCAL share, 12.0 percent, is below the pooled PANCAL mean in Panel A because the share rose between 2000 and the start of the panel: the unweighted mean is 21.2 percent in 2008–2010 and 17.0 percent in 2022–2024. Weighted by the 2000 agricultural workforce, the 2000 share is 30.1 percent and the panel share falls from 41.3 to 33.4 percent between the first and last three panel years; weighted means exceed unweighted means because farm labor is concentrated in high-share zones. The regression-weighted mean of agricultural laborers weights by the 2000 workforce; the first-stage weighted mean $\bar{N}^*$ is the denominator of the substitution rate (Section 4.2).

TABLE 2. The Effect of the Settled Non-Citizen Share on H-2A Employment

	(1) OLS	(2) Two-stage least squares	(3)	(4)
Excluded instruments	—	Budget	Encounters	Both
Panel A. Second stage. Dependent variable: H-2A workers				
PANCAL share (percent)	−2.11	−156.7	−226.5	−178.9
	(1.61)	(61.3)	(65.7)	(60.7)
Substitution rate (headcount)	—	1.02	1.64	1.20
Substitution rate (FTE, NAWS)	—	0.72	1.16	0.85
Panel B. First stage. Dependent variable: PANCAL share				
Border Patrol budget Bartik	—	−0.356	—	−0.304
		(0.093)		(0.092)
Agricultural-season encounter Bartik	—	—	−0.069	−0.047
			(0.012)	(0.010)
Kleibergen–Paap F-statistic	—	14.82	32.45	16.31
Panel C. Reduced form. Dependent variable: H-2A workers				
Border Patrol budget Bartik	—	55.8	—	41.9
		(16.1)		(13.1)
Agricultural-season encounter Bartik	—	—	15.5	12.6
			(4.0)	(3.3)
Anderson–Rubin p-value	—	< 0.001	< 0.001	< 0.001
Hansen J (p-value)	—	—	—	0.167
Observations	11,672	11,672	11,672	11,672

Notes: All regressions include commuting zone fixed effects, year fixed effects, and vegetable share × year interactions, and are weighted by year-2000 commuting zone agricultural employment. Standard errors, clustered at the commuting zone level (740 clusters), in parentheses. Panels B and C are estimated on the column (2)–(4) sample. In column (4), the reduced-form coefficients are from a single regression including both instruments; they are smaller than their single-instrument counterparts in columns (2) and (3) because the two Bartiks share the same exposure shares. The raw correlation between the two Bartiks is 0.997, but after fixed effects and controls the correlation of their residual variation is 0.28. The substitution rate is $|\hat{\delta}|/(0.01 \times \bar{N}^*)$, where $\bar{N}^*$ is the average zone agricultural workforce under the column's first-stage weights (15,426, 13,814, and 14,913 workers in columns (2) to (4)). Section 4.6 reports bootstrap intervals. *Sources:* Department of Labor OFLC disclosure files; American Community Survey via IPUMS; DHS Budget-in-Brief; CBP encounter records.

TABLE 3. Decomposing the OLS–IV Gap: Measurement Error versus Endogeneity

	(1) Coefficient	(2) Share of gap (percent)
Panel A. Estimates. Dependent variable: H-2A workers		
OLS	−2.11	—
	(1.61)	
Errors-in-variables-corrected OLS ($\hat{R} = 0.180$)	−11.7	—
	(9.4)	
Two-stage least squares (Border Patrol budget instrument)	−156.7	—
	(61.3)	
Panel B. Decomposition of the OLS–IV gap		
Total gap (IV minus OLS)	−154.6	100.0
Attributable to measurement error	−9.6	6.2
Attributable to endogeneity	−145.0	93.8
Panel C. Sensitivity of the corrected OLS to the reliability ratio		
Reliability $R = 0.10$	−21.1	12.3
Reliability $\hat{R} = 0.180$ (estimated)	−11.7	6.2
Reliability $R = 0.25$	−8.4	4.1
Reliability $R = 0.35$	−6.0	2.5

Notes: The reliability ratio R is the share of within-commuting-zone variance in the measured PANCAL share attributable to the true share rather than to sampling noise; across bootstrap replications $\hat{R}$ has mean 0.172 and standard deviation 0.019. The standard error of the corrected OLS estimate is from a block bootstrap with 999 replications resampling commuting zones, re-estimating both the coefficient and $\hat{R}$ in each. In Panel B the measurement error component is the corrected OLS estimate minus OLS; the endogeneity component is the IV estimate in Panel A minus the corrected OLS estimate. In Panel C, column (2) reports the share of the total gap that measurement error would explain at each value of R. OLS and IV specifications are as in Table 2.

TABLE 4. Share of H-2A Growth Explained by the PANCAL Decline

Instrument	Rate	Predicted ΔH-2A	Share explained (percent)
Border Patrol budget	1.02	85,243	28.5
Agricultural-season encounters	1.64	137,584	46.0
Both	1.20	100,683	33.7

Notes: The rate is $|\hat{\delta}|/(0.01 \times \bar{N}^*)$ from Table 2, columns (2)-(4). Predicted ΔH-2A is the rate times the decline in the settled non-citizen agricultural workforce, 83,912 workers, summed over the 735 commuting zones observed in both windows; the share divides by the observed H-2A growth of 298,923 over the same zones. Windows are the 2008–2010 and 2022–2024 averages.

TABLE 5. Bootstrap inference for the substitution rate

	Point	Bootstrap SE	Percentile 95 percent interval
Panel A. By unit			
Headcount	1.02	1.04	[0.28, 3.19]
FTE, NAWS 35-week	0.72	0.73	[0.20, 2.26]
FTE, 12-month	0.48	0.49	[0.13, 1.52]
Panel B. Headcount, by first-stage strength of the replication			
All replications (999)			[0.28, 3.19]
KP $F > 5$ (900)			[0.27, 2.30]
KP $F > 10$ (712)			[0.27, 1.95]

Notes: 999 commuting-zone block bootstrap replications of the benchmark specification. The rate divides $|\hat{\delta}|$ by $0.01 \times \bar{N}^*$, with $\bar{N}^*$ held at its full-sample value of 15,426, the average zone workforce under the first-stage weights. Appendix D reports the version that recomputes $\bar{N}^*$ in each replication. FTE conversions use the NAWS 35-week work year (factor 0.707) and a 12-month year (factor 0.476).

TABLE 6. Alternative Exposure Measures

Panel A. Horse races, reduced form. Dependent variable: H-2A workers

	Non-agricultural share	Border distance
Agricultural Bartik (B^{ag})	74.2	74.1
	(26.3)	(23.3)
Alternative Bartik	−82.3	−8.3
	(84.1)	(5.5)

Panel B. Two-stage least squares. Dependent variable: H-2A workers

	Non-agricultural share as control	Border distance as instrument
PANCAL share (percent)	−187.0	−28.0
	(81.6)	(59.8)
Kleibergen-Paap F	12.65	4.67

Notes: The agricultural Bartik is the main instrument, the year-2000 PANCAL share interacted with the Border Patrol budget. The alternative Bartik in Panel A, column (1), interacts the year-2000 settled non-citizen share of low-education non-agricultural employment (high school or less, aged 16 to 64) with the same shifter; the two baseline shares have a correlation of 0.70 across commuting zones. In column (2) it interacts negative log distance to the Mexican border with the same shifter. Each column of Panel A enters the agricultural Bartik and the alternative Bartik in one reduced form, equation (6). Panel B, column (1) adds the contemporaneous non-agricultural settled share as a control to Table 2, column (2), which gives −156.7 (61.3) without it; column (2) instruments the PANCAL share with the distance Bartik alone. Specification, weights, and clustering as in Table 2. Standard errors clustered by commuting zone in parentheses.

TABLE 7. The Livestock Placebo

Panel A. First stage. Dependent variable: livestock PANCAL share (percent)		
	Border Patrol budget	Encounters
Livestock Bartik (B^{lvstk})	−0.57	−0.15
	(0.20)	(0.04)
Panel B. Horse race, equation (7). Dependent variable: H-2A workers		
Crop Bartik (B^{crop})	92.2	
	(29.4)	
Livestock Bartik (B^{lvstk})	−60.6	
	(34.2)	

Notes: Livestock is IPUMS 1990 industry code 11 (livestock production, including dairy, cattle, hogs, and poultry), with the same occupation and settled non-citizen filters as PANCAL. The crop and livestock Bartiks interact the year-2000 settled non-citizen shares of crop and livestock employment with the enforcement shifter; Panel B uses the Border Patrol budget shifter. Livestock cells in the ACS are thin, so the first stage in Panel A is read as a sign and significance check rather than a precise magnitude. Specification, weights, and clustering as in Table 2. Standard errors clustered by commuting zone in parentheses.

TABLE 8. Domestic Agricultural Employment and the Substitution Estimate

	Border Patrol budget	Encounters	Both
Panel A. Domestic employment as the outcome			
PANCAL share (percent)	−381.1	−180.7	−317.3
	(141.5)	(86.2)	(114.6)
Kleibergen-Paap F	14.82	32.45	16.31
Hansen J (p-value)	—	—	0.105
U.S.-born employment as the outcome	—	—	−243.6
			(91.0)
Pre-period joint test, event study (p-value)	0.003 (U.S.-born only: 0.004)		
Panel B. H-2A workers as the outcome, with pre-period domestic growth × year controls			
PANCAL share (percent)	−181.0	−189.1	−184.8
	(97.4)	(72.5)	(81.9)
Kleibergen-Paap F	9.50	27.53	14.04
Hansen J (p-value)	—	—	0.89
Same specification, no control	−156.7	−226.5	−178.9

Notes: Domestic employment is agricultural laborers who are U.S. citizens, born or naturalized. Panel A replaces H-2A workers with domestic employment as the outcome in the specification of Table 2; the U.S.-born row restricts to workers born in the United States. The pre-period test is the joint test that the 2009-2012 coefficients in the domestic event study (Figure 5) equal zero. Panel B adds to the main specification each commuting zone's 2008-2012 domestic employment growth interacted with year dummies. Specification, weights, and clustering as in Table 2.

TABLE 9. Alternative Instruments

	$\hat{\delta}$	Kleibergen-Paap F	Hansen J (p-value)
Panel A. Smooth national shifters in place of the budget			
Consumer price index	−182.8 (63.7)	19.00	—
Federal health spending	−175.4 (59.0)	22.35	—
Linear trend	−168.7 (57.5)	22.36	—
Agricultural-season encounters (non-smooth)	−226.5 (65.7)	32.45	—
Panel B. Supply-shift instruments (baseline share × national PANCAL *decline)*			
Leave-one-out national share, alone	−222.4 (83.9)	16.10	—
National share, alone	−179.9 (60.5)	24.43	—
Leave-one-out share + budget	−177.2 (67.3)	8.40	0.08
Leave-one-out share + encounters	−223.9 (75.5)	17.18	0.92
All three instruments	−180.5 (61.6)	11.93	0.21

Notes: Each row is the specification of Table 2 with the instrument set named. In Panel A the year-2000 PANCAL share is interacted with the log of the named national series in place of the Border Patrol budget. In Panel B the supply-shift instrument interacts the year-2000 PANCAL share with the log of the national PANCAL share, computed excluding the commuting zone's own workers (leave-one-out) or including them. Standard errors clustered by commuting zone in parentheses.

TABLE 10. Rotemberg Weights: Where the Identification Comes From

	Budget	Encounters	Both
Weight in the overidentified estimate	0.68	0.32	—
Weights by period (sum of year weights)			
2008-2017	−23.9	−12.0	−20.1
2018-2024	24.9	13.0	21.1
Three largest year weights	2023, 24, 19	2023, 24, 22	2023, 24, 19
Weights by commuting zone			
California share of identifying weight	0.49	0.37	0.45
Zones with negative weight (of 740)	217	267	227

Notes: Rotemberg decomposition following Goldsmith-Pinkham et al. (2020). The instrument weights are the shares of the overidentified estimate attributable to each just-identified estimate; the weighted just-identified estimates reproduce −156.7, −226.5, and −178.9 exactly. Year weights sum to one within each column; a negative sum over 2008-2017 reflects a national shifter that had not yet moved. The California share is the sum of weights on California commuting zones.

TABLE 11. Controls Progression

	Border Patrol budget		Both instruments	
Controls interacted with year	$\hat{\delta}$	KP F	$\hat{\delta}$	KP F
None	−414.8 (178.5)	11.64	−467.6 (178.7)	17.53
Labor-intensive crop share	−50.7 (96.1)	7.24	−60.3 (107.7)	4.53
Vegetable, orchard, and tobacco shares	−118.1 (84.1)	6.85	−136.7 (88.8)	4.53
Vegetable share (main specification)	−156.7 (61.3)	14.82	−178.9 (60.7)	16.31

Notes: Each row is the specification of Table 2 with the control set named in the first column, all fixed at their 2002 Census of Agriculture values and interacted with year dummies. The labor-intensive crop share is the combined acreage of vegetables, orchards, and berries as a share of cropland; the other rows use the single-crop shares of cropland. Hansen J p-values for the both-instruments column, top to bottom: 0.16, 0.50, 0.16, 0.17. Standard errors clustered by commuting zone in parentheses.

TABLE 12. Specification and Sample Robustness

	$\hat{\delta}$	KP F	J (p)	Rate
Panel A. Levels, both instruments				
Main specification	−178.9 (60.7)	16.31	0.17	1.20
California and weighting				
Excluding California	−346.2 (162.8)	5.55	0.57	—
Excluding three largest California zones	−182.7 (103.5)	8.94	0.93	—
Unweighted	−65.9 (21.7)	28.99	0.52	1.94
Dropping top 1% of H-2A outcomes	−168.2 (94.7)	4.12	0.23	—
Sample and timing				
Crop agriculture only	−211.6 (78.3)	13.06	0.39	2.01
Recent-arrival share control (migration)	−172.9 (56.5)	16.58	0.08	1.06
Recent-arrival share control (years in U.S.)	−181.3 (62.4)	15.34	0.30	1.29
Instruments lagged one year	−179.9 (60.7)	12.02	—	1.41
PANCAL share lagged one year	−193.1 (65.0)	15.35	—	1.29
Wider definitions of the main regressor				
Mexican-born share	−214.3 (77.1)	9.47	0.04	—
Foreign-born share	−216.2 (75.3)	8.55	0.04	—
Panel halves				
2008-2017 only	−69.1 (59.2)	2.43	0.73	—
2018-2024 only	−325.4 (204.3)	1.90	—	—
Panel B. Proportional outcomes, both instruments				
asinh(H-2A workers)	−0.089 (0.038)	16.31	0.33	0.59
ln(H-2A workers), zeros dropped	−0.124 (0.037)	14.79	0.18	0.90

Notes: Each row is the overidentified specification of Table 2, column (4), with the change named. Standard errors clustered by commuting zone in parentheses. Rates are $|\hat{\delta}|/(0.01 \times \bar{N}^*)$ with $\bar{N}^*$ the average zone employment under the first-stage weights of the row's own specification; rates are not reported for rows with a Kleibergen-Paap F below 10, where the first-stage weights that define $\bar{N}^*$ are unstable (Appendix D). In Panel B the implied rate evaluates the proportional effect at the weighted sample-mean H-2A level (987 in the full sample; 1,080 among positive cells) and divides by the specification's $\bar{N}^*$; it is an accounting conversion, not a second estimate. The log specification drops 2,531 of 11,840 commuting-zone-years with zero H-2A workers, 480 of them in commuting zones that never use the program.

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A Data Construction

A.1 H-2A certified positions

The disclosure files include data from fiscal years 2008 through 2024. Because applications are deduplicated by case number, each multi-worksite filing has its certified position(s) counted only once. Worksites were assigned to counties based upon either the county field provided in the files (fiscal year 2017 forward), ZIP codes (fiscal years 2015 and 2016), or city name with manual assignment (fiscal years 2008 through 2014). Once counties were identified, they were mapped to 1990 commuting zones using the crosswalks developed by Dorn (2009) and Autor and Dorn (2013). Geocoding was successful for approximately 97% to 99.6% of positions in the county-field years and approximately 93% to 97.5% of positions in the city-name years. Therefore, a greater percentage of early-year positions are not assigned to a commuting zone.

An application may be certified in full or in part. When an application is partially certified, the number of workers approved is less than what the employer requested, and the certified count on these records represents the reduced amount. Both types of certifications are included in the final outcome, consistent with OFLC's reporting of total certified positions. The constructed series is within 3% of the reported total in all years except fiscal year 2008 when it is 6% greater than the reported figure (approximately 5,100 positions). This difference is approximately equal to the number of positions on partial certifications in the fiscal year 2008 file, indicating that the reported total for fiscal year 2008 likely did not account for partial determinations while the totals for subsequent years did.

Additionally, while OFLC's reported figures include Puerto Rico, the commuting-zone panel does not.

A.2 The settled non-citizen share

Agricultural laborers are identified with 1990 industry codes 10, 11, and 20 (crop production, livestock production, and agricultural services) and 1990 occupation codes 479, 484, and 488 (farm workers, nursery workers, and supervisors). The share is computed with person weights. PUMAs are mapped to commuting zones with the same crosswalks for the 2000 and 2010 PUMA delineations; for the 2020 delineation, I build the crosswalk through Geocorr 2022.

For the exposure share $s_{i,2000}$, I define settled workers in the 2000 census as non-citizens with more than one year in the United States.

A.3 National enforcement series

The budget shifter is the annual Border Patrol budget in 2024 dollars, assembled from Department of Homeland Security Budget-in-Brief documents, Congressional Research Service report RL32562, and appropriations records, and deflated by the CPI-U. The encounter shifter is the number of U.S.-Mexico border encounters that Customs and Border Protection recorded from March through September. The comparison series in Section 5.4 are the CPI-U and real federal health spending, from the Bureau of Labor Statistics and the Bureau of Economic Analysis. Figure A1 plots the series.

A.4 Crop controls

Crop composition data were obtained from the 2002 Census of Agriculture via USDA Quick Stats at the county level; specifically, total cropland and acreage in vegetables, orchards, berries and tobacco. Data were then aggregated into commuting zones using the same crosswalks and each category was converted into a percentage of cropland.

TABLE A1. Variable Definitions and Sources

Variable	Years	Definition	Source
Outcome			
H-2A workers	2008-2024	Certified H-2A positions per commuting zone-year, full and partial certifications	DOL OFLC disclosure files
Main regressor			
PANCAL share	2008-2024	Non-citizen agricultural laborers who did not live abroad in the prior year, as a share of all agricultural laborers (percent)	ACS one-year via IPUMS
Instrument: exposure share			
PANCAL share, 2000	2000	Same construction, with years in the U.S. greater than one replacing the migration filter	2000 census 5 percent sample via IPUMS
Instrument: national shifters			
Border Patrol budget	2008-2024	Log real annual budget, 2024 dollars	DHS Budget-in-Brief; CRS RL32562; CPI-U
Ag-season encounters	2008-2024	Log March-September U.S.-Mexico border encounters	CBP encounter records
Controls			
Crop shares	2002	Vegetable, orchard, berry, and tobacco acreage as shares of cropland	USDA Census of Agriculture
Weights			
Agricultural employment, 2000	2000	Agricultural laborers per commuting zone	2000 census via IPUMS

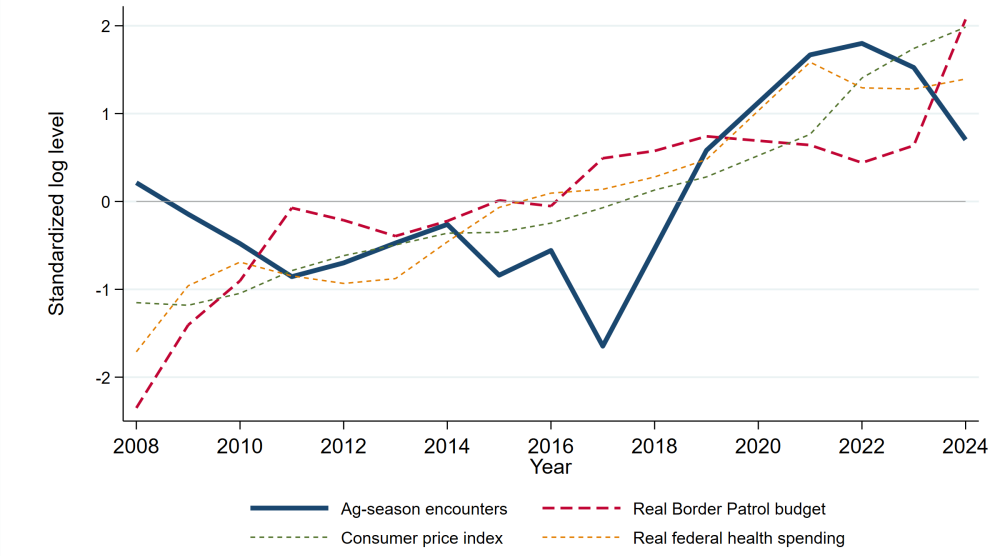


FIGURE A1. National Shifter Series Used to Construct the Instruments

Notes: Each series is the log of the national series, standardized to mean zero and standard deviation one over 2008 to 2024. A linear trend, also used in Section 5.4, is not plotted. *Sources:* DHS Budget-in-Brief; CBP encounter records; Bureau of Labor Statistics; Bureau of Economic Analysis.

B H-2A Contract Duration and the Labor-Time Conversion

The full-time-equivalent substitution rates use the average length of an H-2A contract to calculate the amount of labor time represented by headcount. Figure B1 shows the worker-weighted mean contract duration by year based upon the certification data. Duration was calculated as the requested employment end date minus the start date on each application. Contracts were approximately 6.4 to 6.8 months long through 2011 and have been approximately 5.5 months long since 2013, which covers the period in which the settled workforce declined and H-2A employment increased. Over the 2008 to 2024 period, the worker-weighted mean was 5.71 months. According to the NAWS, all hired crop workers averaged 35 weeks (or 8.08 months) of farm work per year during the six survey cycles that spanned 2007-2020. Therefore, there are two conversion factors: 5.71/8.08 = 0.707 (NAWS benchmark) and 5.71/12 = 0.476 (twelve-month benchmark).

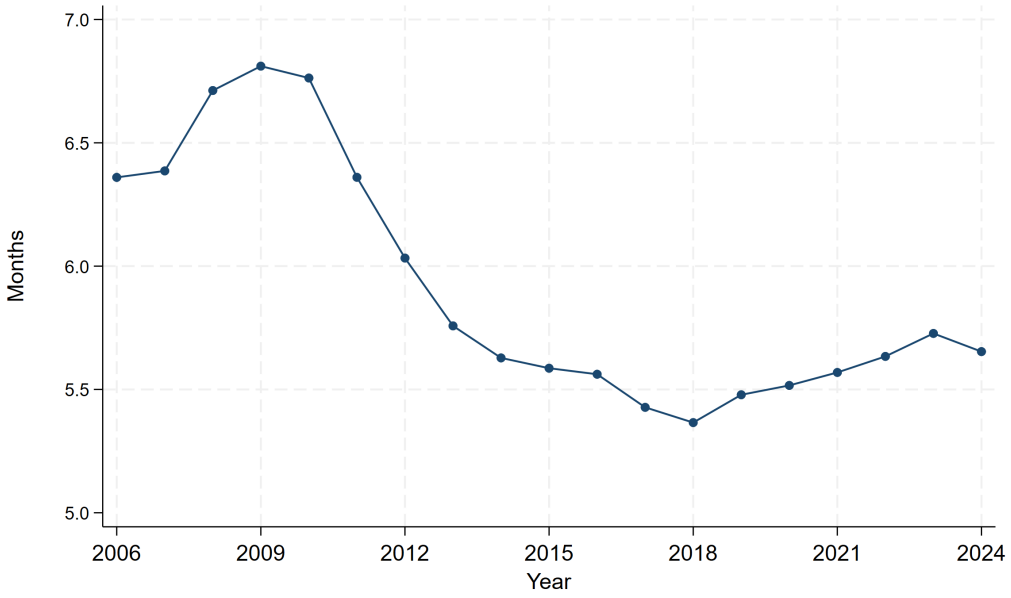


FIGURE B1. Average H-2A Contract Duration by Year

Notes: Worker-weighted mean length of certified H-2A contracts, in months, by fiscal year of certification. Contract length is the requested end date minus the requested start date on each employer application, weighted by the number of workers certified. Applications with missing, non-positive, or implausibly long durations (over 37 months) are dropped. *Sources:* Department of Labor OFLC H-2A disclosure files, 2006–2024.

C Lagged Instruments

Using lagged instruments, I test whether the four-year difference between the decline in the settled share (2015) and the increase in H-2A hiring (2019) is due to a delayed response by H-2A.

I run a regression of H-2A workers in year t on the PANCAL share in year $t-k$, instrumented with the budget Bartik in year $t-k$, for $k = 1$ to 4. If H-2A responds with a delay, then the lagged share will predict H-2A hiring differently than the current share. Lagging drops the first k years, so each row also presents the contemporaneous estimate on the same observations; thus, the only difference between the two columns is timing. Lags that occur in 2020 are omitted since the ACS did not release any standard estimates for that year. All specifications, weights, and clustering are identical to those used in Table 2.

Table C1 displays the results. At all lags, the lagged and contemporaneous estimates are statistically indistinguishable. Both increase as k increases because the shorter samples are shifted toward later years where the effect is larger. A distributed-lag model that includes the share at time t and at time $t-4$ together cannot tell the two apart: the joint first-stage F statistic is 0.3, and neither coefficient is statistically significant on its own. Only their sum is statistically significant (-191.1, $p = 0.004$).

The reason we cannot tell apart the effects of the two shares is the shifter. The log budget has been growing smoothly over time. The budget in year $t-k$ is approximately equal to the budget in year t minus a roughly constant amount. When the log budget is multiplied by the 2000 share, the lagged instrument will be approximately equal to the current instrument minus a zone-specific constant. This zone-specific constant will be removed through the use of commuting zone fixed effects. As a result, both the lagged and current instruments carry almost the same variation. Thus, the design cannot identify the length of the lag.

TABLE C1. Lagged Instruments, Budget Bartik

Lag k	$\hat{\delta}$ at $t-k$	KP F	$\hat{\delta}$ at t, same sample	KP F	Observations
1	−174.1 (70.1)	13.7	−172.3 (67.5)	15.0	10,108
2	−180.2 (69.6)	16.5	−169.4 (66.2)	15.8	9,406
3	−200.3 (76.4)	13.4	−199.7 (85.9)	11.7	8,706
4	−215.9 (87.7)	10.9	−191.1 (69.7)	16.0	7,962

Notes: Each row instruments the PANCAL share at $t-k$ with the budget Bartik at $t-k$ (columns 2-3) and, on the same observations, the share at t with the Bartik at t (columns 4-5). The full-sample contemporaneous estimate is −156.7 (61.3). Standard errors clustered by commuting zone in parentheses. Specification and weights as in Table 2.

D The Substitution Rate: Tail Diagnostics

The bootstrap distribution of the substitution rate is right-skewed, and its right tail has one identifiable source. All results use the benchmark estimate in Table 2, column (2), and the bootstrap of Section 4.6, unless noted.

D.1 Weak draws

Each replication of the simulation will use a different value of $\hat{\delta}$. The rate used in each replication is equal to $|\hat{\delta}|$ divided by $0.01 \times \bar{N}^*$, where $\bar{N}^*$ remains at its full sample size of 15,426. Therefore, the only source of variability in the rate across replications is due to $\hat{\delta}$. Since $\hat{\delta}$ is the ratio of the reduced form coefficient to the first stage coefficient, replications that have a weak first stage will result in large values of $|\hat{\delta}|$. Of the 999 replications, the median Kleibergen-Paap F statistic is 15.3, which is very similar to the 14.8 obtained using the entire sample.

However, 287 replications have an F statistic less than 10. Conditioning on the strength of the first stage removes some of the extreme values (the tail) while leaving the lower end of the distribution unchanged: the 95% confidence intervals are $[0.28, 3.19]$ for all replications; $[0.27, 2.30]$ for replications with $F > 5$; and $[0.27, 1.95]$ for replications with $F > 10$. At the same time, the probability of obtaining a rate less than one increases from 42% to 53%.

D.2 Recomputing the denominator in each replication

In addition to calculating the confidence intervals using the full sample estimate of $\bar{N}^*$, I also ran a second set of bootstrap simulations where each simulation calculated its own set of weights and its own estimate of $\bar{N}^*$. The median rate was close to one (1.32 in headcount; 0.93 in labor time), however the tails were much larger than those obtained previously. The 95% headcount confidence interval was $[0.37, 23.39]$, and in 110 out of 999 replications the sum of the weights was almost zero, making $\bar{N}^*$ undefined. As before, the large tail is due to the same reason. When replications underweight the California core, they will have a weaker first stage. A weaker first stage will increase $|\hat{\delta}|$ and will also cause the weights to be unstable.

Therefore, as $|\hat{\delta}|$ increases, the numerator of the rate increases while the denominator of the rate decreases. There is a negative relationship between $|\hat{\delta}|$ and $\bar{N}^*$ across replications (correlation = 0.50). The smallest possible value of $\bar{N}^*$ that can be computed is 22 workers. If we restrict our analysis to only those replications that have a first-stage F greater than 10, then the confidence interval is reduced to $[0.32, 6.90]$. The instability in the estimates is

caused by the resampling process and not by the full sample estimate, since the full sample estimate of $\bar{N}^*$ is based on a strong first stage ($F = 14.8$).

D.3 No single zone drives the coefficient

The leave-one-zone-out jackknife estimates the benchmark and $\bar{N}^*$ 740 times, each time removing one zone. The median rate is 1.02, and 739 of the 740 rates lie within $[0.85, 1.86]$. The only rate outside this range occurs when the Salinas commuting zone is removed: $\bar{N}^*$ decreases from 15,426 to 2,182 workers, the first-stage F decreases to 7.7, and the rate increases to 9.6, the same weak-first-stage mechanism as in the bootstrap. Although the rate changes substantially, the coefficient itself does not change much: when $\bar{N}^*$ is held at its full-sample value, all 740 rates, including the one without Salinas, lie within $[0.84, 1.36]$. Therefore, the large change in the rate is due to the denominator and not to $\hat{\delta}$.

D.4 Which places

In each bootstrap replication, a zone may be selected zero, one, or more times. I calculate the correlation between the number of times a zone is selected and the rate in that replication. A negative correlation means that the rate is lower than average when the zone is selected more often than average, which indicates that the absence of this zone increases the rate. Conversely, a positive correlation means that the rate is higher than average when the zone is selected more often than average. The zones whose inclusion lowers the rate, and whose combined exclusion produces the tail, form the California core. The California core includes the Fresno-Tulare San Joaquin Valley zone (which alone accounts for 5.7 percent of national agricultural employment in 2000), greater Los Angeles, the Salinas Valley, Kern County, and Merced-Stanislaus. These zones are characterized by high baseline exposure, with PANCAL shares of 60 to 72 percent in 2000, large size, and low H-2A intensity (0.04 to 0.14 H-2A workers per agricultural worker, although the Salinas Valley has a higher intensity of 0.50). The zones whose inclusion raises the rate are those with high H-2A intensity. Examples include the Yakima Valley, the Columbia Basin, and Adams-Grant in Washington, as well as Palm Beach and Polk-Highlands in Florida. These zones have 0.8 to 2.2 H-2A workers per agricultural worker, an order of magnitude more than the California core.

The national rate is, in effect, an average across all of the commuting zones. However, there are large differences in how much H-2A workers are used in each zone. California has a large settled workforce, relatively low use of H-2A workers, and much of the weight, so it anchors the national rate close to one. Draws that underweight California reduce the variation available to identify the effect, weaken the first stage, and allow the H-2A-intensive

regions of Washington and Florida to drive the results. A check on the full sample supports this. If I drop the three largest California commuting zones (Fresno-Tulare, greater Los Angeles, and the Salinas Valley), the coefficient in the overidentified specification remains virtually identical (−182.7 against −178.9), but the identifying workforce drops from 14,913 to 3,953 workers and the first stage weakens to an F of 8.9, so no rate is reported. Therefore, the tail is due to the geography of the denominator, not to an unstable coefficient.

I also test whether the tail is driven by the largest H-2A cells. Dropping the top 1 percent of commuting-zone-year H-2A outcomes gives a coefficient of −168.2 (94.7), so the largest cells are not driving the estimate. Because this test is weakly identified (first-stage F of 4.1), it is read as a comparison of point estimates only.

E Measurement Diagnostics

The PANCAL measure excludes non-citizens who lived abroad in the prior year. Three diagnostics show that the measure does not count H-2A workers.

E.1 Group quarters

The American Community Survey (ACS) began including group quarters such as labor camps, worker dormitories and employer housing in 2006; therefore, guest workers would show up in this category if they were present in the survey. Figure E1 shows the percentage of agricultural workers who live in group quarters by group. Over the period of 2006 to 2024, the percentage of settled non-citizen agricultural workers who lived in group quarters was constant at approximately 5%. However, over the same time period, the percentage of recently arrived non-citizen agricultural workers who lived in group quarters averaged 34% and increased from 28% prior to 2018 to 46% after 2018 as H-2A employment increased. Only 1.9% of domestic workers lived in group quarters. This increase in group quarter residency occurred only among recent arrivals and not among settled workers. Since the number of unweighted respondents in each cell of the recent arrival group is small (between 187 and 601 per year), the annual series is variable. Thus removing recent arrivals from the settled workers definition ensures removal of H-2A visa holders to some extend.

E.2 Seasonality

The ACS considers a worker to be an agricultural laborer only if they are working in agriculture at the time of the interview; therefore, the ACS may underestimate the number of seasonal

workers who are interviewed during their off-season. If settled non-citizen workers were either more or less seasonal than domestic workers, then this would cause a bias in the share.

According to monthly data from the Current Population Survey, there is no difference between the two groups. Both groups have their highest levels in July and their lowest levels in March. The gap between the peak and trough for each group was 29.8 and 32.8 index points respectively over the period 2008-2024 (Figure E2).

E.3 The crop controls

The vegetable share of cropland is used as the crop control in the main specification. Two figures illustrate why it is the composition of the crops grown by certified H-2A workers that is important, and what it removes. Figure E3 displays the crop composition of certified H-2A workers from 2011 to 2019, which are the years that the disclosure files list a crop. In 2011, the vegetable share was 16 percent. By 2016, it had increased to 32 percent and remained above a quarter through 2019. Conversely, the general/unspecified category decreased from 20 percent to less than 10 percent before increasing again. Berries increased from 3 to 9 percent; while tobacco, apples and nursery stock remained constant.

Therefore, the expansion of the program is concentrated in vegetables, which is also the crop group in which settled non-citizen workers are most likely employed. If a commuting zone had vegetable acreage at baseline, then it would have experienced an increase in H-2A hiring over this time regardless of whether its settled workforce changed. The interaction of the vegetable share with year dummies accounts for this effect. Figure E4 presents the reduced-form event study results with and without the vegetable controls. The coefficients 2017 onward are roughly double when the vegetable controls are removed. The coefficients prior to 2017 are nearly zero regardless of the controls.

Thus, the crop-demand confound inflates the magnitude of the response after 2017 but does not cause either the timing or pattern of responses before 2017.

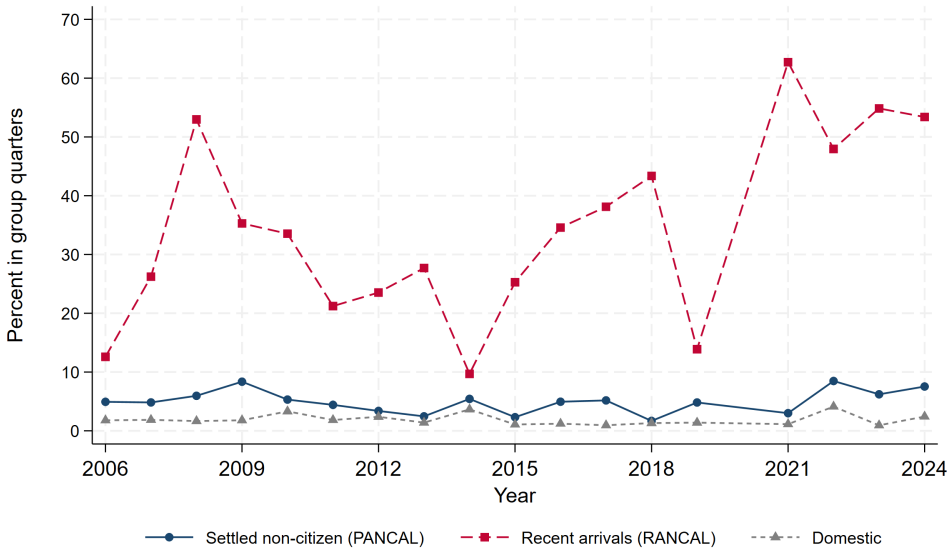


FIGURE E1. Share of Agricultural Laborers Living in Group Quarters, by Group

Notes: Person-weighted share of agricultural laborers in group quarters (IPUMS gq codes 3 and 4), by year and group. PANCAL is settled non-citizens; recent arrivals are non-citizens who lived abroad one year earlier; domestic is citizens. *Source:* ACS one-year files via IPUMS, 2006-2024.

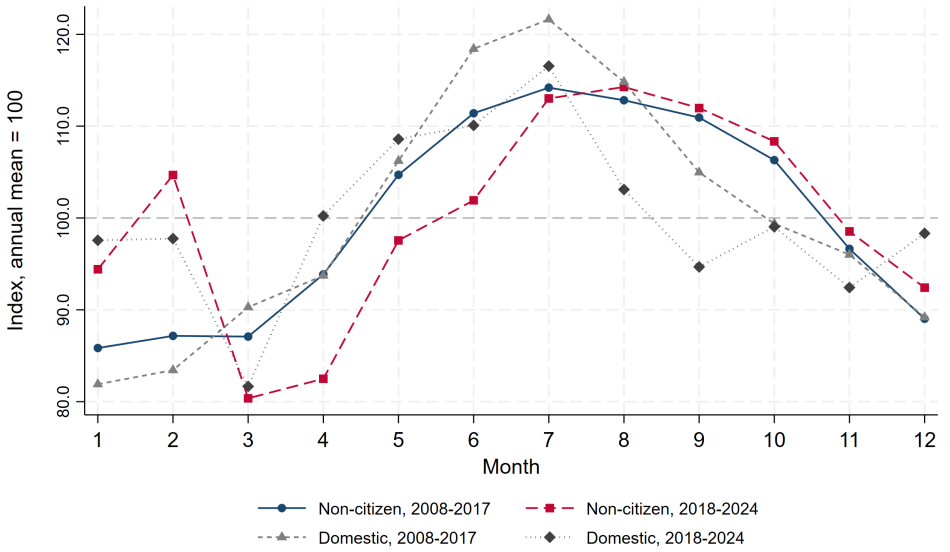


FIGURE E2. Seasonal Profile of Agricultural Laborers, CPS Monthly

Notes: Monthly employment of agricultural laborers indexed to the annual mean of 100, by citizenship and era. Same industry and occupation definitions as the ACS frame. Roughly 250 to 350 respondents per month. *Source:* CPS Basic Monthly files via IPUMS, 2008-2024.

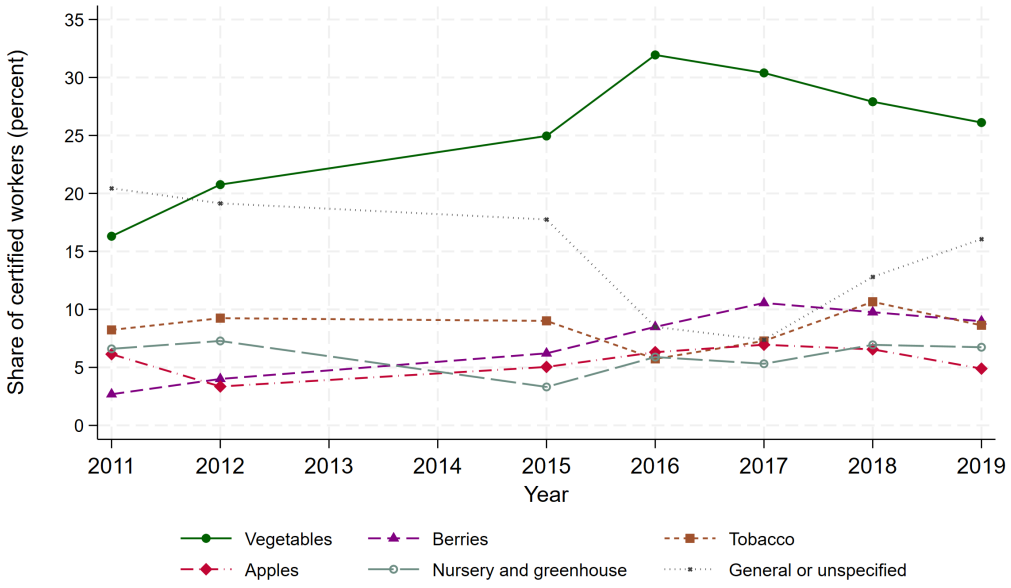


FIGURE E3. Crop Composition of Certified H-2A Workers

Notes: Worker-weighted shares of certified H-2A positions by crop, from the OFLC disclosure files. Crop information is unavailable for 2013 and 2014 and is not reported after 2019. Vegetables include melons, following the USDA-NASS classification. Only the six largest crop groups are shown; the general or unspecified category covers applications without a crop designation.

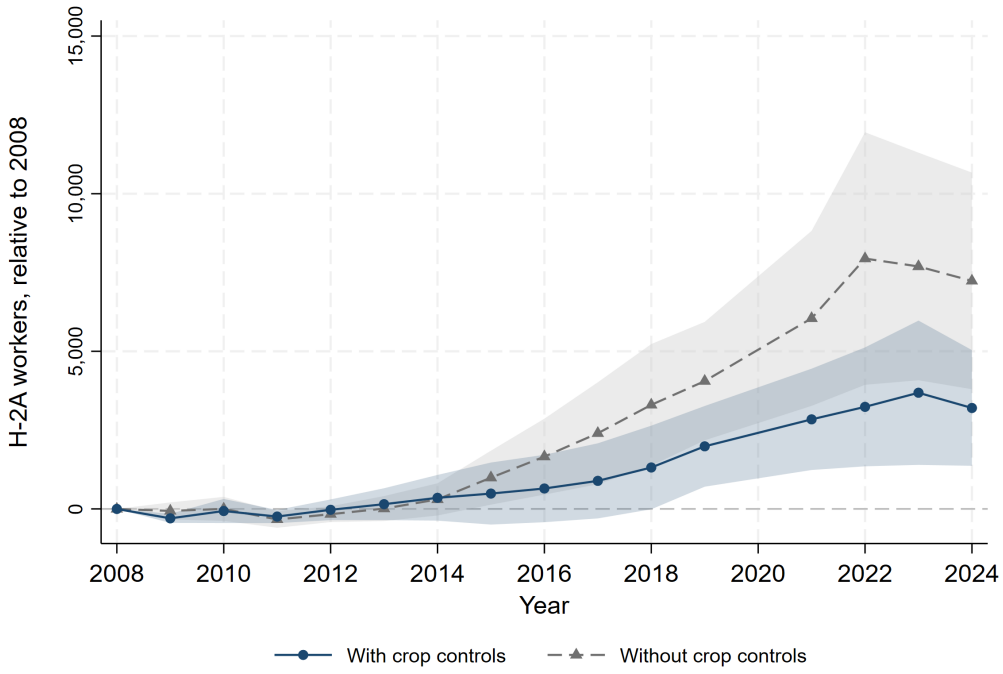


FIGURE E4. Reduced-Form Event Study With and Without Crop Controls

Notes: Coefficients on the year-2000 PANCAL share interacted with year dummies, with H-2A workers as the outcome and 2008 as the reference year, estimated with and without the vegetable share × year controls of the main specification. 740 commuting zones, weighted by year-2000 agricultural employment. Shaded bands are 95 percent confidence intervals clustered at the commuting zone level.